

# Risk-Aware Last-Mile Routing under Persistent Disruptions: Tail-Risk Planning and Selective Dynamic Recourse

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## Abstract

Last-mile delivery systems are increasingly exposed to disruptions that are temporally persistent rather than independent across time. We study a stochastic capacitated vehicle routing problem with time windows (CVRPTW) in which an exogenous Hawkes process represents clustered disruptions, while route-dependent schedule propagation determines how those disruptions translate into service delay. The framework combines risk-aware Adaptive Large Neighbourhood Search (ALNS), learning-augmented operator selection, and causal Dynamic Data-Driven Application Systems (DDDAS) recourse. Experiments use nine Athens-derived operational instances under controlled stress scenarios, with common random numbers and matched stochastic search budgets.

Increasing disruption persistence substantially raises upper-tail tardiness even when the continuous-time mean environmental travel-time multiplier is held approximately constant: for an unchanged benchmark route,  $\text{CVaR}_{0.95}$  tardiness increases from 60.74 to 113.24 between the low- and high-persistence regimes. The strongest algorithmic result is obtained from explicit tail-risk optimization. Across all nine operational days, CVaR-oriented ALNS improves the aligned tail-risk objective relative to mean-oriented ALNS by 8.61 on average and reduces  $\text{CVaR}_{0.95}$  tardiness by 7.28, with positive paired gains on all nine days. By contrast, feed-forward and recurrent reinforcement-learning operator selectors show no robust

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advantage over adaptive ALNS or a uniform-selection control under matched online budgets and zero-shot transfer.

Sequential DDDAS recourse is selective rather than uniformly beneficial. Among 523 recourse attempts, 87 produce route modifications; all improve the local conditional risk objective at decision time, yet realized composite cost improves in only 18 of the 65 runs containing a route change. These results distinguish persistent environmental risk, ex-ante tail-aware planning, and ex-post adaptive recourse, and indicate that the most reproducible gains arise from explicit CVaR-based preoperational routing rather than from additional learning or continual rerouting.

*Keywords:* Last-mile logistics, CVRPTW, Persistent disruptions, Conditional Value-at-Risk, Hawkes processes, Adaptive large neighbourhood search, Dynamic routing

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## 1. Introduction

Urban last-mile logistics increasingly operate in regimes where service reliability is dominated not by average conditions, but by rare yet persistent disruptions that propagate through tightly coupled delivery schedules. Congestion waves, local incidents, and the depletion of schedule slack interact over time, producing delay cascades whose impact may far exceed that predicted by classical memoryless uncertainty models. Travel-time conditions in urban routing can vary systematically over time, and recent reviews of time-dependent and stochastic dynamic routing emphasize that temporal structure, information arrival, and real-time reoptimization materially affect route quality (Ritzinger et al., 2016; Zhang and Van Woensel, 2023; Adamo et al., 2024; Mardešić et al., 2024). In such environments, routing plans that are optimal—or even robust—under static or weakly stochastic assumptions may exhibit substantial fragility once executed sequentially over time.

The Capacitated Vehicle Routing Problem with Time Windows (CVRPTW) is a cornerstone of transportation science and operations research, with more than five decades of methodological development (Solomon, 1987; Toth and Vigo, 2002; Laporte, 2009). State-of-the-art exact, heuristic, and metaheuristic approaches—most notably Large Neighbourhood Search (LNS) and Adaptive Large Neighbourhood Search (ALNS)—are capable of producing high-quality solutions for large-scale deterministic instances (Bräysy and Gendreau, 2005; Røpke and Pisinger,

2006; Christiaens and Vanden Berghe, 2020). Modern industrial solvers such as Google OR-Tools further illustrate the practical scalability of deterministic vehicle-routing optimization under nominal travel-time assumptions (Google OR-Tools Team, 2026). In their standard deterministic use, however, such formulations do not explicitly represent temporal persistence in the disruption process or the resulting exposure of a route to clustered periods of degraded travel conditions.

A substantial literature has demonstrated that ignoring travel-time uncertainty can lead to considerable degradation of operational performance. Stochastic, robust, and distributionally robust routing formulations address this problem through scenario-based, probabilistic, or ambiguity-set representations of uncertainty (Ritzinger et al., 2016; Zhang et al., 2021; Mardešić et al., 2024). Risk-sensitive criteria provide an additional mechanism for controlling service unreliability. In particular, Conditional Value-at-Risk (CVaR) offers a coherent way to target the upper tail of tardiness rather than optimizing expected performance alone (Rockafellar and Uryasev, 2000). Yet temporal dependence and persistence remain less developed than marginal travel-time uncertainty itself: scenario-based formulations can represent heterogeneous and even correlated outcomes without necessarily describing how disruption episodes cluster, decay, and become observable during sequential route execution.

In parallel, learning-augmented optimization has gained prominence in vehicle routing. Reinforcement learning (RL) has been applied to end-to-end route construction (Nazari et al., 2018; Kool et al., 2019), while recent surveys document a broader shift toward learning-augmented routing and hybrid combinations of learned policies with classical optimization or improvement heuristics (Bogyrbayeva et al., 2024; Zhou et al., 2025; Wang et al., 2025). These hybrid approaches are attractive because learning can guide expensive search decisions while combinatorial operators preserve the structural constraints of the routing problem. However, an important question remains insufficiently resolved: whether learned operator selection provides an operational advantage once search effort, stochastic evaluation budget, offline training cost, and generalization to unseen instances are controlled. This question becomes particularly relevant when the optimization criterion is not mean performance but tail risk, and when the state available to a routing policy evolves with the observed disruption history.

From a systems perspective, two distinct temporal mechanisms must be separated. First, disruption conditions may themselves exhibit *exogenous*

*persistence*: recent events contain information about the likelihood of further events even though routing decisions do not alter the disruption-generating process. Second, delays propagate *through the delivery schedule*: a slow arc changes later departure times, consumes time-window slack, and can transform subsequent travel-time realizations into tardiness. The second mechanism can therefore amplify the operational consequences of the first without generating feedback from the route to the disruption process. This distinction is important because a route may be exposed to the same external disruption environment as another route while experiencing different consequences simply because its arc departures occur at different times.

We represent the first mechanism using an exogenous self-exciting Hawkes process (Hawkes, 1971; Bacry et al., 2015). Recent disruption events generate a decaying excitation state, which is mapped to an *Operational Memory Index* (OMI) and then to a multiplicative travel-time effect. The Hawkes process is common to the operating environment and is not affected by realized routing delays. For the exponential kernel used here, the excitation admits a finite-dimensional Markov representation; accordingly, the system is history-dependent relative to a conventional routing state that omits this excitation variable, but it becomes Markovian once the environmental memory state is included. Route dependence then arises because different routing decisions induce different departure times and hence different exposure to the same evolving OMI trajectory.

Against this background, the present study asks three related but distinct questions. First, how does disruption persistence affect mean and tail routing risk when differences in average environmental impact are controlled? Second, does explicitly optimizing tail risk through CVaR provide a reproducible advantage over mean-risk optimization under a common stochastic evaluation protocol? Third, once a risk-aware preoperational route is available, how much additional value is provided by causal online recourse based only on information observed up to each decision epoch? A complementary learning experiment asks whether feed-forward or recurrent RL-based operator selection improves upon the underlying adaptive search once computational budgets and generalization are controlled.

To address these questions, we formulate a stochastic CVRPTW with two-dimensional vehicle capacity and a common risk-sensitive objective,

$$J_\rho(x) = \lambda_D D(x) + \lambda_T \rho(Z_x) + \lambda_W O_W(x) + \lambda_V O_V(x),$$

where  $D(x)$  denotes route distance,  $Z_x$  is cumulative tardiness under stochas-

tic execution,  $O_W(x)$  and  $O_V(x)$  denote weight and volume capacity violations, and  $\rho$  is either the mean or  $\text{CVaR}_\alpha$ . The same objective structure is used across the principal search and learning comparisons. We implement a genuine adaptive ALNS and compare mean-oriented and CVaR-oriented search under common random numbers and matched candidate–scenario evaluation budgets. Feed-forward and gated recurrent unit (GRU) policies are evaluated as learned destroy–repair operator selectors rather than as replacements for the combinatorial solver.

The framework is then extended to sequential operation through a Dynamic Data-Driven Application Systems (DDDAS) recourse layer. At each eligible decision epoch, only the disruption events and travel-time information observed before departure are available. When the OMI enters a prospectively specified high-memory episode, the system may reconsider the residual route of the affected vehicle. Future disruption events are generated conditionally from the observed Hawkes state, already traversed arcs retain their observed travel times, and unobserved travel times remain stochastic. The resulting recourse problem is therefore causal and local to the residual vehicle suffix. Online search is explicitly bounded, so a disruption trigger should be interpreted as a request to *reconsider* the incumbent route rather than as an instruction to reroute.

The empirical analysis uses multi-day last-mile delivery instances derived from real operations in Athens, Greece (Vrani et al., 2025). The operational data provide the customer, demand, spatial, and travel-time structure of the routing instances, while moderately tightened time windows and the Hawkes disruption process define controlled stress scenarios used to expose reliability differences among methods. The disruption process should therefore be interpreted as an experimental persistence model rather than as a Hawkes process calibrated from observed Athens incident arrivals. Nine operational days are used for the principal cross-instance evaluation, and three prospectively selected representative days are examined more deeply for sequential DDDAS behavior.

The experiments reveal a differentiated picture. Increasing disruption persistence substantially increases tail tardiness risk even under an auxiliary control that matches the continuous-time mean environmental travel-time multiplier across persistence regimes. At the planning stage, CVaR-oriented ALNS consistently improves the tail-oriented objective relative to mean-oriented ALNS across all nine operational days, indicating that explicit tail-risk optimization provides a reproducible benefit beyond optimiz-

ing expected tardiness alone. By contrast, neither feed-forward nor recurrent learned operator selection exhibits a robust advantage over the adaptive search or a uniform operator-selection control once search budgets, validation, and transfer to other days are considered. Thus, recurrent state augmentation is not assumed to be beneficial merely because the environment contains temporal persistence.

The sequential DDDAS experiments provide a further distinction between *decision-time conditional risk* and *realized ex-post performance*. Across the confirmatory sequential experiment, recourse was triggered frequently but actual route modification was selective. Every executed route modification improved the local conditional search objective relative to its incumbent residual route, yet these ex-ante conditional improvements did not translate systematically into lower realized composite cost on the subsequently observed trajectory. The online mechanism is therefore best interpreted as bounded, state-dependent recourse under uncertainty rather than as a guaranteed realized-cost improvement procedure. This result also illustrates why evaluating an online risk-sensitive controller solely by the realized outcome of a single stochastic trajectory can obscure the distinction between the objective optimized when the decision is made and the realization that subsequently occurs.

The contributions of this work are fourfold.

1. We develop a persistent-disruption formulation of the stochastic CVRPTW that explicitly separates an exogenous Hawkes disruption process from route-dependent schedule propagation and exposure.
2. We provide a controlled assessment of temporal persistence and tail risk, including a mean-impact-matched experiment that isolates changes in disruption clustering from changes in average environmental travel-time impact.
3. We establish a common-budget comparison of mean-oriented ALNS, CVaR-oriented ALNS, and learning-augmented operator-selection policies, including feed-forward and recurrent alternatives and zero-shot transfer across operational days.
4. We implement and evaluate a causal, computationally bounded DDDAS recourse mechanism in which decisions depend only on information available at the decision epoch, thereby quantifying both the potential and the limitations of online risk-sensitive adaptation.

Taken together, the results suggest that resilience under persistent disruption

tion should not be identified with a single algorithmic mechanism. Explicit tail-risk optimization provides the clearest and most reproducible improvement at the preoperational planning stage, whereas learning-based search control and online recourse provide additional adaptive capabilities whose value depends on the information, computational budget, and realized trajectory. This separation between environmental persistence, risk-aware planning, and sequential recourse provides a more transparent basis for designing data-driven decision support for disruption-prone last-mile logistics.

## 2. Related Work

### *2.1. Vehicle routing with time windows under uncertainty*

The vehicle routing problem with time windows (VRPTW) has developed from the classical deterministic formulation of Solomon (1987) into a broad family of stochastic, robust, time-dependent, and dynamic routing models (Toth and Vigo, 2002; Laporte, 2009; Bräysy and Gendreau, 2005). Recent reviews stress that uncertainty is not characterized only by its marginal magnitude: the timing of information, temporal variation of travel conditions, and the possibility of reoptimization are central modelling choices (Ritzinger et al., 2016; Zhang and Van Woensel, 2023; Mardešić et al., 2024; Adamo et al., 2024). Our setting keeps the daily customer set fixed before dispatch, while a common exogenous disruption environment evolves during execution. Dynamic adaptation therefore enters through causal residual-route reconsideration, not through new customer arrivals.

### *2.2. Risk-averse routing and tail performance metrics*

Expectation-based objectives can understate operational exposure when a small number of severe delays dominate service penalties. Risk-averse routing therefore uses chance constraints, robust or distributionally robust formulations, quantiles, and coherent tail-risk measures such as Conditional Value-at-Risk (CVaR) (Rockafellar and Uryasev, 2000). Distributionally robust vehicle-routing models have, for example, used CVaR approximations to control deadline violations when the underlying distribution is uncertain (Zhang et al., 2021). The contribution here is not CVaR itself, but a controlled test of whether explicit CVaR optimization remains beneficial when uncertainty is modulated by a persistent latent disruption state and when average environmental impact is held fixed.

### *2.3. Metaheuristics and adaptive large neighbourhood search*

Adaptive Large Neighbourhood Search (ALNS) is a well-established framework for rich routing problems (R pke and Pisinger, 2006). Its defining feature is adaptive operator selection based on observed search performance rather than a fixed neighbourhood schedule. A recent systematic review of 211 VRP studies provides a unified taxonomy of removal and insertion operators and reports strong performance for foresight-based insertions such as regret insertion (Voigt, 2025). ALNS also remains relevant in explicitly dynamic VRPTW variants; for example, Wang et al. (2024) develop an ALNS reoptimization framework for a multi-depot dynamic VRPTW. These results motivate using a genuinely adaptive ALNS as the main combinatorial benchmark here. Mean- and CVaR-oriented variants share the same adaptive machinery, neighbourhoods, and stochastic budget, isolating the optimized risk functional.

### *2.4. Learning-augmented routing and reinforcement learning*

Learned methods now span end-to-end route construction and augmentation of classical search. Early policy-gradient and attention models demonstrated that neural policies can construct competitive routing solutions (Nazari et al., 2018; Kool et al., 2019). A 2024 survey synthesizes machine-learning approaches to VRPs (Bogyrbayeva et al., 2024), and a 2025 review organizes recent routing work into construction- and improvement-based learning paradigms (Zhou et al., 2025). Hybrid RL–ALNS methods continue to appear, including PPO-guided operator control for VRPTW (Wang et al., 2025). The present learning component follows this hybrid paradigm: feed-forward and recurrent policies select destroy operators, while feasibility, repair, stochastic scoring, and acceptance remain in the common ALNS engine. This permits an incremental test against both adaptive ALNS and uniform operator selection, with offline training cost reported separately from the matched online search budget.

### *2.5. Temporal dependence, self-excitation, and dynamic control*

Time-dependent and stochastic dynamic routing already establish that route quality can depend strongly on when travel conditions change and when new information is available (Zhang and Van Woensel, 2023; Adamo et al., 2024). Self-exciting point processes provide a complementary mechanism for representing event clustering. Hawkes’ original construction (Hawkes, 1971)

has since been used broadly for clustered event processes, and a recent transportation example models road accidents in Rome with a spatio-temporal Hawkes process (Alaimo Di Loro et al., 2025). Here the Hawkes process is deliberately an *experimental exogenous disruption model*: it is not calibrated from Athens incident arrivals and routing decisions do not feed back into its intensity. Its excitation state determines a common Operational Memory Index (OMI), while different routes experience different exposure because they depart onto arcs at different times.

Dynamic Data-Driven Application Systems (DDDAS) provide a natural systems concept for the execution stage because observations acquired during operation can update the state used for subsequent decisions (Darema, 2004). In our implementation, this principle is causal and local: only events and completed travel times observed before a decision epoch condition the residual problem, and only the unexecuted suffix of the affected vehicle can be revised.

### 2.6. Positioning of this work

The gap addressed here lies at the intersection of these mature streams. We study a CVRPTW in which (i) disruption events exhibit explicit temporal persistence through an exogenous excitation state, (ii) persistence is separated from average environmental impact through matched controls, (iii) mean- and CVaR-oriented routing are compared using the same adaptive ALNS and common stochastic budgets, (iv) learned operator selection is evaluated as an incremental augmentation rather than assumed to dominate adaptive search, and (v) online adaptation is implemented as causal, bounded residual-route recourse. Accordingly, we do not claim that Hawkes processes uniquely represent urban disruption or that RL is intrinsically superior; the experiments identify which additional mechanisms are supported under the common confirmatory protocol.

## 3. Problem Formulation

We consider a stochastic capacitated vehicle routing problem with soft time windows in which vehicles operate under an exogenous disruption environment with temporal persistence. The formulation separates three components that play different roles during execution: (i) baseline origin–destination travel-time uncertainty, (ii) an exogenous self-exciting disruption process, and (iii) route-dependent schedule propagation. Routing decisions

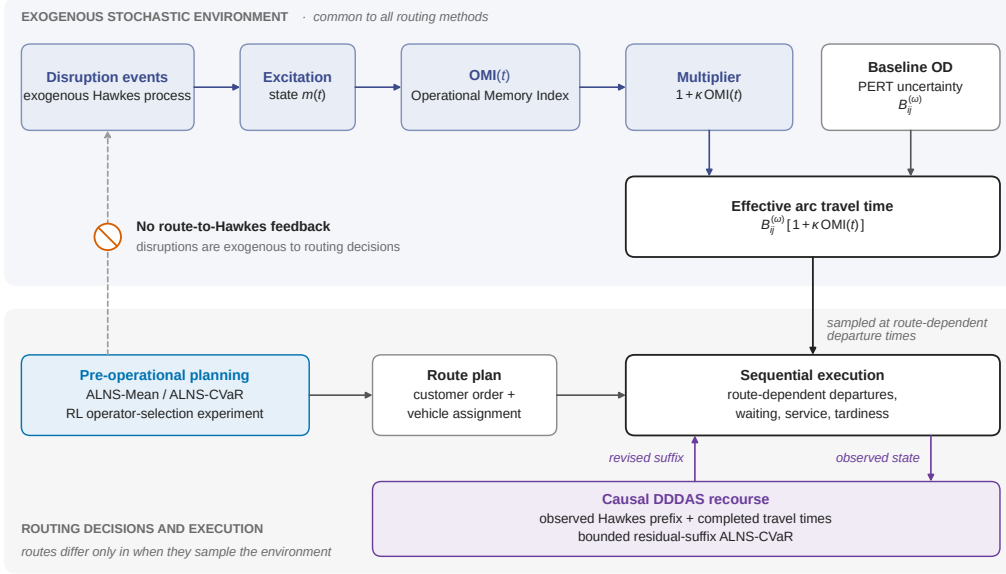


Figure 1: **Conceptual architecture of persistent-risk routing.** The disruption process is exogenous to routing. Baseline OD uncertainty and the Hawkes-derived OMI jointly determine effective arc travel times, while route choices determine the departure times at which the common environmental state is sampled. Preoperational risk-aware planning is followed by sequential execution. The DDDAS layer uses only the observed disruption prefix and completed travel times to reconsider the unexecuted vehicle suffix; the revised suffix then returns to execution. No route-to-Hawkes feedback is assumed (crossed dashed arrow).

affect when a vehicle is exposed to the evolving disruption state, but they do not affect the disruption-generating process itself.

### 3.1. Network, routes, demands, and capacity

Let  $G = (V, A)$  be a directed graph with depot 0 and customer set  $\mathcal{C} = \{1, \dots, n\}$ , so that  $V = \{0\} \cup \mathcal{C}$ . The fleet is denoted by  $\mathcal{K} = \{1, \dots, K\}$ . Each customer  $i \in \mathcal{C}$  is characterized by a weight demand  $w_i$ , a volume demand  $v_i$ , a service duration  $s_i$ , and a service window  $[a_i, b_i]$ .

The upper endpoint  $b_i$  is treated as a service target rather than as a hard feasibility cutoff: vehicles may serve a customer after  $b_i$ , but the resulting tardiness is penalized. Early arrival is permitted and induces waiting until  $a_i$ .

A routing solution can be represented either by binary arc variables  $x_{ij}^k \in$

$\{0, 1\}$  or equivalently by a collection of vehicle routes

$$x = \{r_1, \dots, r_K\}, \quad r_k = (0, i_{k,1}, \dots, i_{k,n_k}, 0).$$

Each customer is assigned to exactly one route.

Let  $Q_k^W$  and  $Q_k^V$  denote the weight and volume capacities of vehicle  $k$ . The corresponding route loads are

$$L_k^W(x) = \sum_{i \in r_k \cap \mathcal{C}} w_i, \quad L_k^V(x) = \sum_{i \in r_k \cap \mathcal{C}} v_i.$$

Capacity excesses are

$$O_W(x) = \sum_{k \in \mathcal{K}} [L_k^W(x) - Q_k^W]_+, \quad (1)$$

$$O_V(x) = \sum_{k \in \mathcal{K}} [L_k^V(x) - Q_k^V]_+, \quad (2)$$

where  $[z]_+ = \max\{z, 0\}$ . The implemented destroy–repair search enforces both capacity dimensions when constructing admissible routes; consequently, these violation terms are zero for all reported feasible solutions and serve as safeguards in the canonical scoring function.

The deterministic routing distance is

$$D(x) = \sum_{k \in \mathcal{K}} \sum_{\ell=0}^{n_k} d_{i_{k,\ell}, i_{k,\ell+1}}, \quad (3)$$

where  $i_{k,0} = i_{k,n_k+1} = 0$  and  $d_{ij}$  is the observed OD distance.

### 3.2. Baseline travel-time uncertainty

For each OD pair  $(i, j)$ , the operational data provide optimistic, most-likely, and pessimistic travel-time estimates,

$$t_{ij}^{\text{opt}}, \quad t_{ij}^{\text{ml}}, \quad t_{ij}^{\text{pes}}.$$

These triplets are retained unchanged in the source data. For stochastic sampling only, occasional ordering inconsistencies are regularized as

$$L_{ij} = \min \{t_{ij}^{\text{opt}}, t_{ij}^{\text{ml}}, t_{ij}^{\text{pes}}\}, \quad M_{ij} = t_{ij}^{\text{ml}}, \quad U_{ij} = \max \{t_{ij}^{\text{opt}}, t_{ij}^{\text{ml}}, t_{ij}^{\text{pes}}\}. \quad (4)$$

The baseline stochastic travel time  $B_{ij}^{(\omega)}$  is sampled from a modified PERT distribution on  $[L_{ij}, U_{ij}]$  with mode  $M_{ij}$  and shape parameter  $\lambda_{\text{PERT}} = 4$ . For  $U_{ij} > L_{ij}$ , the corresponding beta-shape parameters are

$$a_{ij} = 1 + \lambda_{\text{PERT}} \frac{M_{ij} - L_{ij}}{U_{ij} - L_{ij}}, \quad (5)$$

$$b_{ij} = 1 + \lambda_{\text{PERT}} \frac{U_{ij} - M_{ij}}{U_{ij} - L_{ij}}. \quad (6)$$

Degenerate triplets with  $L_{ij} = U_{ij}$  yield a deterministic baseline time.

The regularization in Eq. (4) is used only to define a valid stochastic sampling distribution; it does not modify the raw OD data. The nominal deterministic routing benchmark uses the corresponding nominal most-likely travel-time information rather than stochastic PERT draws.

### 3.3. Exogenous persistent disruption process

Temporal clustering of disruptions is represented by a univariate Hawkes process (Hawkes, 1971; Bacry et al., 2015). Let  $\{\tau_q\}_{q \geq 1}$  denote disruption-event times. With an exponential kernel, the conditional event intensity is

$$\lambda_H(t) = \mu + \xi \sum_{\tau_q < t} \exp[-\beta(t - \tau_q)], \quad (7)$$

where  $\mu > 0$  is the baseline event rate,  $\beta > 0$  is the decay rate, and  $\xi > 0$  is the excitation amplitude.

We parameterize the excitation amplitude as

$$\xi = \eta\beta, \quad 0 \leq \eta < 1, \quad (8)$$

so that  $\eta = \xi/\beta$  is the Hawkes branching ratio. For constant  $\mu$ , the stationary mean event intensity is

$$\bar{\lambda} = \frac{\mu}{1 - \eta}. \quad (9)$$

Importantly, the Hawkes process is *exogenous to routing*. The event process is generated independently of route choices, realized travel times, and realized tardiness. A delayed vehicle therefore does not increase future Hawkes intensity. The interaction between the route and the disruption process occurs only through the times at which the route samples the external disruption state.

For the exponential kernel, define the excitation state

$$m(t) = \xi \sum_{\tau_q < t} \exp[-\beta(t - \tau_q)]. \quad (10)$$

Between events,  $m(t)$  decays exponentially, and at a disruption event it jumps by  $\xi$ . Hence the exponential Hawkes model admits a finite-dimensional Markov embedding through  $m(t)$ . The process is therefore history-dependent relative to a conventional routing state that omits this variable, but it is not intrinsically non-Markovian once the excitation state is included.

#### 3.4. Operational Memory Index and route exposure

We summarize the instantaneous contribution of past events to the current disruption intensity through the Operational Memory Index (OMI),

$$\text{OMI}(t) = \frac{m(t)}{\mu + m(t)} \in [0, 1]. \quad (11)$$

An OMI close to zero indicates that the current intensity is dominated by the baseline component  $\mu$ , whereas values close to one indicate that recent events dominate the instantaneous intensity.

OMI is an *environment-level* quantity: for a fixed Hawkes realization, the same function  $\text{OMI}(t)$  is available to every routing method. Methods do not suppress or modify OMI. They can, however, experience different *route-dependent OMI exposure* because different routes depart on different arcs at different times.

#### 3.5. Effective travel times and sequential execution

For scenario  $\omega$ , the effective travel time on arc  $(i, j)$  when that arc is started at time  $t$  is

$$T_{ij}^{(\omega)}(t) = B_{ij}^{(\omega)} \left[ 1 + \kappa \text{OMI}^{(\omega)}(t) \right], \quad (12)$$

where  $\kappa \geq 0$  controls the operational impact of the disruption state.

Equation (12) defines exposure at *arc departure*. Once an arc has been started, its travel-time multiplier is fixed; events that occur during traversal do not retrospectively modify that arc. They become observable only at a subsequent decision epoch.

Consider route

$$r_k = (0, i_1, \dots, i_{n_k}, 0)$$

under scenario  $\omega$ . Let  $A_\ell^{(\omega)}$  denote arrival time at customer  $i_\ell$ ,  $S_\ell^{(\omega)}$  its service-start time, and  $D_\ell^{(\omega)}$  the departure time after service. The execution recursion is

$$D_0^{(\omega)} = 0, \quad (13)$$

$$A_\ell^{(\omega)} = D_{\ell-1}^{(\omega)} + T_{i_{\ell-1}, i_\ell}^{(\omega)} \left( D_{\ell-1}^{(\omega)} \right), \quad (14)$$

$$S_\ell^{(\omega)} = \max \left\{ A_\ell^{(\omega)}, a_{i_\ell} \right\}, \quad (15)$$

$$D_\ell^{(\omega)} = S_\ell^{(\omega)} + s_{i_\ell}, \quad \ell = 1, \dots, n_k. \quad (16)$$

Customer tardiness is measured at service start,

$$Z_{i_\ell}^{(\omega)}(x) = \left[ S_\ell^{(\omega)} - b_{i_\ell} \right]_+, \quad (17)$$

and total scenario tardiness is

$$Z_x^{(\omega)} = \sum_{k \in \mathcal{K}} \sum_{\ell=1}^{n_k} Z_{i_\ell}^{(\omega)}(x). \quad (18)$$

These recursions create route-level path dependence even though the disruption process itself is exogenous. An early travel-time realization can alter later departure times, deplete time-window slack, and thereby change the OMI values sampled by subsequent arcs. Two methods exposed to the same Hawkes trajectory can therefore experience different tardiness because their routes induce different temporal exposure profiles.

### 3.6. Mean- and tail-risk objectives

For a routing solution  $x$ , we use the canonical risk-sensitive objective

$$J_\rho(x) = \lambda_D D(x) + \lambda_T \rho(Z_x) + \lambda_W O_W(x) + \lambda_V O_V(x), \quad (19)$$

where  $\rho$  is either expected total tardiness,

$$\rho(Z_x) = \mathbb{E}[Z_x],$$

or Conditional Value-at-Risk,

$$\rho(Z_x) = \text{CVaR}_\alpha(Z_x).$$

All principal experiments use  $\alpha = 0.95$ . The same objective structure and penalty convention are used for ALNS and the learning-augmented variants; only the risk functional  $\rho$  changes between mean-oriented and CVaR-oriented experiments.

Given a frozen scenario bank  $\Omega = \{\omega_1, \dots, \omega_M\}$ , the sample mean is

$$\hat{\mathbb{E}}_\Omega[Z_x] = \frac{1}{M} \sum_{m=1}^M Z_x^{(\omega_m)}. \quad (20)$$

Empirical CVaR is evaluated as the expected shortfall of the empirical distribution. Equivalently,

$$\widehat{\text{CVaR}}_{\alpha, \Omega}(Z_x) = \min_{\tau \in \mathbb{R}} \left\{ \tau + \frac{1}{(1 - \alpha)M} \sum_{m=1}^M [Z_x^{(\omega_m)} - \tau]_+ \right\}. \quad (21)$$

This definition correctly handles finite samples and ties at the empirical quantile.

Candidate and incumbent routes are compared on the *same* frozen scenario bank whenever they compete within a search step. This common-random-number construction reduces Monte Carlo comparison noise and prevents differences between algorithms from being driven by different stochastic realizations.

### 3.7. Optimization questions

The formulation leads to two distinct decision problems.

The first is a *preoperational planning problem*: before execution begins, find a route set that performs well under the full stochastic disruption environment. We compare mean-oriented and CVaR-oriented versions of Eq. (19).

The second is an *online residual-routing problem*: once execution is under way, determine whether the unexecuted suffix of an affected vehicle should be modified using only information observed up to the current decision epoch. This second problem is conditional, causal, and computationally bounded. It is developed in Section 4.

## 4. Methodology

The computational framework separates preoperational route optimization from online recourse. The same stochastic execution model is used

throughout, but the information available to the optimizer differs between the two stages. Preoperational methods evaluate complete routes against frozen scenario banks. The DDDAS layer instead constructs a conditional scenario bank from the information available at the current execution time and optimizes only the remaining suffix of the triggered vehicle.

#### 4.1. Scenario banks and common random numbers

A scenario bank contains two stochastic components: a realization of the exogenous Hawkes event process and a matrix of baseline OD travel-time draws. For a fixed route and scenario, sequential execution follows Eqs. (12)–(18).

Separate frozen scenario banks are used for training, validation, search, and final testing. Within a comparison, all competing methods are evaluated on the same final test scenarios. During local search, incumbent and candidate routes are also evaluated against the same search bank. This common-random-number design is central to the fairness protocol because route differences are evaluated under identical realizations of the external disruption environment.

#### 4.2. Adaptive Large Neighbourhood Search

Adaptive Large Neighbourhood Search (ALNS) provides the combinatorial search backbone (Røpke and Pisinger, 2006). Starting from a feasible route set  $x_k$ , each iteration applies a destroy operator followed by a repair procedure to construct a candidate  $\tilde{x}_k$ .

The destroy set contains random, worst-contribution, and Shaw-related removal operators. At each non-null proposal, the number of removed customers is generated from a fraction uniformly sampled between 0.10 and 0.15 of the current customer set, with at least two removals. Candidate reinsertion uses a randomized regret-2 procedure with feasible-position screening under both weight and volume capacities. High-regret choices are rank-biased rather than selected deterministically, allowing controlled diversification while preserving feasibility.

A candidate is scored through the stochastic objective

$$\widehat{J}_\rho(x; \Omega)$$

on the current frozen scenario bank  $\Omega$ . Search uses simulated-annealing acceptance: an improving candidate is accepted, while a non-improving candidate may be accepted according to the current temperature. The initial

temperature is 5 and the cooling factor is 0.995 for each evaluated non-null candidate.

The baseline ALNS is genuinely adaptive. Operator weights are initialized equally and updated every 25 operator decisions using

$$w_a^{\text{new}} = 0.8 w_a^{\text{old}} + 0.2 \max \left\{ \frac{R_a}{N_a}, 10^{-6} \right\}, \quad (22)$$

where  $R_a$  is accumulated operator reward and  $N_a$  is its number of uses in the adaptation window. Rewards distinguish three outcomes: creation of a global best solution, improvement of the current solution, and accepted non-improving movement. The corresponding rewards are 8, 4, and 1, respectively.

This adaptive baseline is used for both the mean-oriented and CVaR-oriented search. Consequently, the comparison

ALNS-Mean   versus   ALNS-CVaR

changes the risk functional being optimized rather than replacing a non-adaptive heuristic with an adaptive one.

#### 4.3. Learning-augmented operator selection

We additionally investigate whether learned search control can improve upon the adaptive ALNS baseline. The learning component does not construct routes directly. Instead, it selects the destroy neighbourhood used to propose the next ALNS move, while the same feasibility checks, repair mechanism, stochastic objective, and acceptance rule remain in force.

At search step  $k$ , the policy observes the nine-dimensional state

$$s_k = \left( \bar{D}_k, \bar{T}_k^{\text{nom}}, u_k^W, u_k^V, \bar{Z}_k, \sigma_{Z,k}, p_k^{\text{OT}}, \bar{E}_k^{\text{OMI}}, E_k^{\text{OMI,max}} \right), \quad (23)$$

where the components are, respectively, normalized route distance, normalized nominal tardiness, maximum weight utilization, maximum volume utilization, normalized Monte Carlo mean tardiness, normalized Monte Carlo tardiness standard deviation, mean on-time service fraction, mean route OMI exposure, and maximum route OMI exposure.

The last two quantities are route-exposure statistics and should not be interpreted as changes to the environmental OMI itself. They summarize the OMI values encountered at the route's actual arc-departure times.

Two policy classes are considered. The feed-forward policy conditions directly on  $s_k$ . The recurrent policy uses a gated recurrent unit (GRU),

$$h_k = \text{GRU}_\theta(s_k, h_{k-1}), \quad a_k \sim \pi_\theta(\cdot \mid h_k). \quad (24)$$

Only the newly observed state  $s_k$  is processed at step  $k$  together with the previous hidden state  $h_{k-1}$ ; the full preceding sequence is not processed again.

Learning is aligned with exactly the same stochastic objective used by the search. If  $x_k$  is the incumbent route set and  $\tilde{x}_k$  the proposed candidate, both evaluated on scenario bank  $\Omega_b$ , the immediate learning signal is

$$r_k = \hat{J}_\rho(x_k; \Omega_b) - \hat{J}_\rho(\tilde{x}_k; \Omega_b). \quad (25)$$

Thus a positive reward corresponds to a candidate with a lower objective on the same stochastic realizations. Simulated-annealing acceptance is handled separately: it determines whether the candidate becomes the next search state, but it does not redefine the reward of a rejected proposal.

Feed-forward and recurrent policies are evaluated against both the adaptive ALNS baseline and a uniform operator-selection control. Offline policy-training cost is accounted for separately from the online search budget, while final method-to-method search comparisons use matched candidate–scenario evaluation budgets. This design allows the learning component to be assessed as an empirical augmentation rather than assuming that recurrence must be beneficial because the environment contains temporal persistence.

#### 4.4. Causal DDDAS recourse

The Dynamic Data-Driven Application Systems (DDDAS) component (Darema, 2004) operates during route execution. Unlike the preoperational optimizers, it has access only to information that would have been observed by the decision time.

A decision epoch occurs immediately before a vehicle starts a new arc, after any waiting and service at its current node have been completed. At departure time  $t_0$ , the observable disruption history is

$$\mathcal{H}_{t_0} = \{\tau_q : \tau_q < t_0\};$$

events at or after  $t_0$  are not available to the controller. Similarly, baseline travel times are regarded as observed only for arcs whose traversal has already been completed.

For the exponential Hawkes model, the observed prefix determines the current excitation state

$$s_H(t_0) = \sum_{\tau_q < t_0} \exp[-\beta(t_0 - \tau_q)], \quad m(t_0) = \xi s_H(t_0). \quad (26)$$

Future Hawkes events can therefore be generated conditionally from the pair  $(t_0, s_H(t_0))$  without access to the unobserved future realization.

Conditional scenario generation at  $t_0$  follows three rules:

1. the observed Hawkes prefix is held fixed and future events are sampled conditionally from the state in Eq. (26);
2. baseline travel times for already completed OD traversals retain their observed realized values;
3. baseline travel times for unobserved OD pairs are newly sampled from the PERT model in Section 3.2.

No latent future events and no unobserved entries of the originally realized OD matrix are supplied to the replanner.

#### 4.5. Memory-triggered episodes

Online reconsideration is governed by a prospectively specified OMI trigger. The controller is initially armed. The first eligible departure satisfying

$$\text{OMI}(t_0) \geq 0.55 \quad (27)$$

opens a high-memory episode. The controller is rearmed only after a subsequent observed departure has  $\text{OMI}(t) < 0.55$ ; no additional cooldown interval is imposed.

Within an episode, each vehicle can receive at most one recourse attempt, and a vehicle is eligible only if at least two customers remain unserved on its current route. A trigger therefore denotes *route reconsideration*, not mandatory rerouting.

The online optimizer cannot change the already executed portion of the route. At the decision epoch, the following elements are fixed: the current vehicle node, its executed prefix, its customer assignment, and the final return to the depot. The decision variable is only the ordering of the remaining customers assigned to the triggered vehicle.

#### 4.6. Conditional local CVaR replanning

Let  $r_{k,t_0}^{\text{inc}}$  denote the incumbent residual suffix of triggered vehicle  $k$  at time  $t_0$ . A conditional scenario bank  $\Omega_{t_0}$  is generated from the observed information described above. The local online objective uses the same distance and tardiness weights as the preoperational formulation, with  $\rho = \text{CVaR}_{0.95}$ .

The incumbent and every candidate suffix are evaluated on the same conditional scenario bank. Define

$$G_{t_0} = \hat{J}_{\text{CVaR}}(r_{k,t_0}^{\text{inc}}; \Omega_{t_0}) - \hat{J}_{\text{CVaR}}(r_{k,t_0}^{\text{sel}}; \Omega_{t_0}), \quad (28)$$

where  $r_{k,t_0}^{\text{sel}}$  is the best suffix returned by the bounded search. The suffix is replaced only when the conditional search objective improves; hence an executed rerouting decision satisfies  $G_{t_0} > 0$  up to numerical tolerance.

This is deliberately a *local triggered-vehicle* recourse problem. It should not be interpreted as exact minimization of the conditional CVaR of the entire fleet. Final performance, however, is always measured by replaying the complete fleet execution under the same realized environmental trajectory.

#### 4.7. Bounded online search

Each conditional replanning attempt uses 100 conditional scenarios. The ALNS search seeks to evaluate up to 250 distinct non-null candidate suffixes but is subject to a hard limit of 25,000 operator decisions. Null proposals count against the operator-decision budget but require no scenario evaluation.

If the target number of candidate evaluations is reached first, search terminates normally. If the operator-decision limit is reached first, the search terminates at the cap and returns the best admissible suffix found up to that point. Consequently, if  $C \leq 250$  non-null candidates are actually evaluated, the conditional scenario-evaluation cost of that attempt is

$$N_{\text{eval}} = 100(1 + C), \quad (29)$$

including the incumbent evaluation.

The stopping rule makes online recourse computationally bounded even in residual states for which the destroy–repair neighbourhood produces very few distinct feasible alternatives.

#### 4.8. Sequential execution algorithm

Algorithm 1 summarizes the causal execution protocol.

The static and DDDAS executions used for paired evaluation share the same realized Hawkes path and baseline travel-time realization. Thus differences between them arise from route decisions rather than from different environmental draws.

#### 4.9. Methodological separation of *ex-ante* and *ex-post* quantities

A final distinction is essential for interpreting online performance. The quantity in Eq. (28) is an *ex-ante conditional* comparison based on a distribution of possible future continuations at the decision epoch. By contrast, realized route cost is computed after execution from the single future trajectory that actually occurs,

$$J_{\text{realized}} = D_{\text{realized}} + 2Z_{\text{realized}}, \quad (30)$$

for the confirmatory sequential experiment.

Accordingly,

$$G_{t_0} > 0$$

does not imply that the selected suffix must produce a lower  $J_{\text{realized}}$  on every subsequent realization. The former is a conditional risk statement; the latter is one sample from the conditional future distribution. Maintaining this separation prevents decision-time risk improvement from being conflated with guaranteed *ex-post* performance.

## 5. Experimental Design and Evaluation

The experimental design was constructed to separate four questions that are frequently conflated in dynamic stochastic routing: (i) whether temporal persistence changes routing risk, (ii) whether explicit tail-risk optimization improves preoperational routing, (iii) whether learned operator selection improves upon adaptive search under a comparable computational budget, and (iv) whether causal online recourse provides additional value once route execution has begun.

All stochastic comparisons use common random numbers whenever methods are evaluated against one another. Search, validation, transfer, and sequential recourse experiments were frozen before the corresponding confirmatory comparisons were inspected. Results from earlier exploratory implementations are not included in the analyses reported below.

Table 1: Structural characteristics of the nine Athens-derived daily instances. The final column reports the fraction of OD travel-time triplets requiring ordering regularization for stochastic PERT sampling. The raw source values themselves are not modified.

| Day | Customers | Weight (kg) | Volume (m <sup>3</sup> ) | PERT-reg. arcs (%) |
|-----|-----------|-------------|--------------------------|--------------------|
| 1   | 78        | 1880.498    | 9.0200                   | 7.61               |
| 2   | 63        | 1394.549    | 5.2281                   | 23.14              |
| 3   | 67        | 687.633     | 4.0536                   | 8.89               |
| 4   | 69        | 1689.368    | 4.7584                   | 0.62               |
| 5   | 75        | 1270.167    | 4.4493                   | 1.04               |
| 6   | 77        | 1301.434    | 4.5925                   | 2.93               |
| 7   | 66        | 1034.561    | 3.6932                   | 0.25               |
| 8   | 74        | 1095.370    | 3.2131                   | 0.22               |
| 9   | 84        | 1301.537    | 5.3159                   | 0.10               |

### 5.1. Athens-derived operational instances

The experiments use nine daily last-mile delivery instances derived from the Athens operational data described by Vrani et al. (2025). The source data provide the spatial structure, customer orders, weight and volume demands, service information, and OD travel-time descriptors used to construct the routing instances.

The nine days contain between 63 and 84 customer orders, excluding the depot, and exhibit substantial heterogeneity in daily demand and in the consistency of the OD travel-time triplets. Table 1 summarizes the principal structural characteristics used in the present study.

The empirical routing data and the experimental disruption mechanism play different roles. The Athens data define the operational base instances, whereas the tightened service windows and Hawkes disruption process define controlled stress scenarios. Accordingly, the experiments should be interpreted as *Athens-derived real operational instances under controlled stress scenarios*, not as a reconstruction of historical disruption events in Athens.

The vehicle fleet is fixed at six vehicles. Weight and volume capacities are experimental design parameters established before the comparative analyses; they should not be interpreted as observed capacities of a specific Athens fleet. Capacity feasibility is enforced simultaneously in both dimensions.

For the deterministic OR-Tools computations, continuous demand quantities are mapped to integer units without truncating positive demands.

Weight is scaled by  $10^3$  and volume by  $10^6$  before integer optimization. Stochastic evaluation is subsequently carried out in the original physical units.

The controlled stress protocol uses service-window widths of 180 minutes. Its purpose is not to reproduce a specific historical operating day exactly, but to provide a common reliability environment in which differences among mean-oriented, tail-oriented, learning-augmented, and dynamically adaptive routing policies can be exposed.

### 5.2. Preoperational route initialization

To avoid method-specific advantages arising from different starting solutions, route initialization is performed independently of the stochastic search methods.

For the nine-day cross-instance experiment, a common OR-Tools initialization protocol was frozen before comparative stochastic optimization. The protocol tests a fixed ordered collection of standard first-solution strategies, including local cheapest insertion, path-most-constrained arc, savings, global-cheapest arc, and path-cheapest arc, combined with guided local search. Each strategy is allocated the same time allowance, and the first feasible solution generated by the frozen protocol is retained.

This fallback mechanism is applied uniformly across days rather than selecting the best initializer retrospectively for each instance. In particular, different strategies may provide the first feasible route on different days without constituting instance-specific tuning.

The earlier source-day learning validation on Day 1 uses its independently frozen Day 1 initialization. The subsequent nine-day experiment uses the common cross-day initialization protocol described above. Numerical results from these two stages are therefore interpreted within their respective experimental designs and are not pooled as if they arose from an identical initial route.

### 5.3. Nominal stochastic configuration

The nominal persistent-disruption configuration uses

$$\beta = 0.60, \quad \eta = 0.35, \quad \mu = 0.01, \quad \kappa = 0.90. \quad (31)$$

With the parameterization  $\xi = \eta\beta$ , the nominal excitation amplitude is  $\xi = 0.21$ , and the stationary mean Hawkes event intensity is

$$\bar{\lambda} = \frac{0.01}{1 - 0.35} \simeq 0.01538. \quad (32)$$

The modified PERT model uses  $\lambda_{\text{PERT}} = 4$ , and all tail-risk experiments use

$$\alpha = 0.95. \quad (33)$$

The canonical objective weights are

$$\lambda_D = 1, \quad \lambda_T = 2, \quad \lambda_W = \lambda_V = 10. \quad (34)$$

Because weight and volume feasibility are enforced by the route-construction mechanism, the last two terms remain zero for reported feasible solutions.

#### 5.4. *Frozen stochastic evaluation banks*

Four conceptually distinct scenario-bank roles are used:

$$\Omega_{\text{train}}, \quad \Omega_{\text{val}}, \quad \Omega_{\text{search}}, \quad \Omega_{\text{test}}.$$

The corresponding baseline configuration uses

$$|\Omega_{\text{train}}| = |\Omega_{\text{val}}| = |\Omega_{\text{search}}| = 100, \quad |\Omega_{\text{test}}| = 1000. \quad (35)$$

Training scenarios are used only when fitting learned policies. Validation scenarios support policy selection without using the final test bank. Search scenarios are used to compare candidate and incumbent routes during optimization. The test bank is held out from search and is used only for final performance evaluation.

Within any paired comparison, the same test scenarios are supplied to all methods. Candidate and incumbent solutions within ALNS or learned search also share the same search bank. Thus stochastic differences between methods are paired at the scenario level rather than being confounded with independently generated disruption realizations.

#### 5.5. *Search-budget matching*

A central objective of the experimental design is to distinguish algorithmic performance from computational expenditure. For the principal ALNS and frozen learned-policy comparisons, online search is therefore matched by the number of candidate-route-scenario evaluations.

Each search uses 100 search scenarios and evaluates up to 250 non-null candidate route sets in addition to the incumbent. The nominal stochastic evaluation budget is consequently

$$B_{\text{search}} = 100(1 + 250) = 25,100 \quad (36)$$

scenario evaluations.

The same search budget is applied to adaptive ALNS and to the feed-forward and GRU-guided variants. Runtime is reported as a complementary quantity, but is not used as the sole fairness criterion because stochastic objective evaluation and neighbourhood generation have method-dependent computational structure.

Offline RL training is *not* included inside  $B_{\text{search}}$ . Instead, it is reported separately as an additional up-front computational cost. In the source-day experiments, fitting one policy requires approximately  $2.3 \times 10^5$  scenario evaluations, or roughly an order of magnitude more stochastic evaluations than one frozen-policy search. This distinction prevents the cost of policy learning from being hidden inside a nominally equal online-search comparison.

### 5.6. Source-day learning experiment

The first learning experiment is conducted on Day 1 and compares six risk-search combinations:

$$\begin{array}{lll} \text{ALNS-Mean,} & \text{RL-FF-Mean,} & \text{RL-GRU-Mean,} \\ \text{ALNS-CVaR,} & \text{RL-FF-CVaR,} & \text{RL-GRU-CVaR.} \end{array}$$

Here “FF” denotes the feed-forward operator-selection policy and “GRU” denotes the recurrent policy in Eq. (24).

Ten independent paired repetitions are used for final source-day evaluation. Within each repetition, competing methods share the relevant frozen stochastic banks. Learned policies are trained and validated before final test evaluation; the 1000-scenario final test bank is not used for policy fitting or model selection.

A uniform operator-selection policy is evaluated as an additional control. It uses the same search machinery and stochastic budget but samples among the available destroy operators uniformly rather than through learned or adaptive selection. This control is important because it distinguishes the value of a trained policy from improvements that could arise simply through stochastic operator variation.

The source-day analysis therefore addresses two different questions: whether learned policies outperform the genuine adaptive ALNS baseline, and whether any observed learned-policy advantage exceeds that of a simple uniform operator selector.

### 5.7. Zero-shot cross-day policy transfer

Generalization is evaluated without target-day retraining. Policies are trained only on Day 1 and then transferred directly to Days 2–9. No target-day learning, fine-tuning, or target-specific adjustment of policy parameters is permitted.

The transfer experiment uses 40 frozen source-policy checkpoints, corresponding to four checkpoints from each of ten source training repetitions. Every checkpoint is evaluated on the eight target days using the same target-day route initializations and stochastic evaluation protocol as the corresponding uniform-policy control.

The primary transfer contrasts compare learned and uniform operator selection for both feed-forward and recurrent policies under the mean- and CVaR-oriented objectives. The target day, rather than an individual checkpoint or stochastic run, is the cross-instance unit of analysis. Four prespecified primary transfer contrasts are corrected jointly using the Holm procedure.

This design tests zero-shot transportability of the learned search policy rather than its ability to be retrained on each new instance.

### 5.8. Persistence experiment

A separate set of experiments isolates the effect of temporal persistence. The Hawkes branching ratio is varied over

$$\eta \in \{0, 0.15, 0.35, 0.55, 0.70\}. \quad (37)$$

A direct change in  $\eta$  would simultaneously alter event clustering and the stationary mean event rate. To prevent this confounding, the first control fixes the stationary mean intensity at its nominal value,  $\bar{\lambda}$ , by setting

$$\mu_\eta = \bar{\lambda}(1 - \eta). \quad (38)$$

Thus  $\eta$  changes the temporal clustering of the disruption process without changing its stationary mean number of events.

For this persistence experiment, the operational memory transformation that maps realized events into OMI retains the nominal decay and excitation parameters. This ensures that the  $\eta = 0$  generator provides a meaningful Poisson-like no-self-excitation comparison while the route can still experience nonzero disruption exposure when external events occur.

Before routing comparisons, 10,000 environmental trajectories are generated for each persistence regime. These simulations are used only to characterize the experimental manipulation—including event-count dispersion, no-event frequency, event timing, and time-averaged OMI—rather than to tune routing algorithms.

#### 5.9. Mean-impact-matched persistence control

Matching event intensity does not by itself guarantee equal mean operational impact because changes in clustering can alter the time profile of OMI. We therefore introduce a second, stricter control that matches the *continuous-time mean environmental travel-time multiplier*

$$\mathbb{E}_t [1 + \kappa_\eta \text{OMI}_\eta(t)] . \quad (39)$$

The nominal target is

$$\mathbb{E}_t[M(t)] = 1.052489.$$

The resulting prospectively computed impact coefficients are

|               |        |        |        |        |         |      |
|---------------|--------|--------|--------|--------|---------|------|
| $\eta$        | 0      | 0.15   | 0.35   | 0.55   | 0.70    | (40) |
| $\kappa_\eta$ | 0.6953 | 0.7646 | 0.9000 | 1.1418 | 1.4816. |      |

This control matches the continuous-time environmental mean multiplier, not the mean multiplier sampled by any particular route. Route-sampled exposure remains route-dependent because methods depart on arcs at different times.

The confirmatory persistence-routing experiment is performed on Day 1 using ten paired repetitions per persistence regime. Candidate search uses 100 scenarios and final evaluation uses 1000 scenarios. No search parameter is retuned as  $\eta$  changes.

The experiment evaluates both the direct change in routing tail risk as persistence increases and whether persistence systematically changes the relative advantage of CVaR-oriented over mean-oriented ALNS.

#### 5.10. Nine-day risk-aware routing experiment

Cross-instance robustness of explicit tail-risk optimization is assessed on all nine Athens-derived days under the nominal disruption configuration. Three paired optimization repetitions are performed on each day using the common cross-day initialization protocol.

The principal comparison is

$$\text{ALNS-Mean} \quad \text{versus} \quad \text{ALNS-CVaR}.$$

Both methods use the same adaptive ALNS engine, neighbourhoods, search length, scenario-bank size, and final test bank. The comparison therefore isolates the effect of changing the optimized risk functional from expected tardiness to  $\text{CVaR}_{0.95}$ .

For each repetition, candidate search is based on 100 stochastic scenarios and final evaluation on 1000 held-out scenarios. The three optimization replications are first averaged within each operational day. Consequently, the cross-instance inferential unit is the *day*, giving  $n = 9$  independent day-level contrasts rather than treating the 27 optimization runs as independent instances.

The two prespecified primary contrasts are

$$\Delta J_{\text{CVaR}} = J_{\text{CVaR}}(\text{ALNS-Mean}) - J_{\text{CVaR}}(\text{ALNS-CVaR}), \quad (41)$$

$$\Delta T_{\text{tail}} = \text{CVaR}_{0.95}(Z_{\text{ALNS-Mean}}) - \text{CVaR}_{0.95}(Z_{\text{ALNS-CVaR}}). \quad (42)$$

Positive values favor CVaR-oriented search.

Across-day uncertainty is summarized using bootstrap confidence intervals over the nine day-level contrasts. Because the direction of the paired contrast is prespecified, an exact two-sided sign-flip test is additionally used as a distribution-free check of cross-day consistency.

### 5.11. *Confirmatory sequential DDDAS experiment*

The online DDDAS experiment is deliberately separated from the nine-day preoperational analysis. Its purpose is not to establish cross-instance superiority over the entire dataset, but to examine the mechanics and realized consequences of causal sequential recourse in greater depth.

Three representative instances—Days 2, 5, and 7—were selected prospectively from the nine-day set using structural information only, before the confirmatory sequential outcomes were inspected. Day 1 was excluded from this selection because it had already been used extensively during development and runtime validation.

For each selected day, the three independently generated ALNS-CVaR routes from the nine-day experiment are frozen as preoperational starting solutions. Twenty realized environmental trajectories are then generated per day. The same realized Hawkes and baseline travel-time trajectory is replayed

across the three preoperational route replicates for a given day–path pair, producing

$$3 \text{ days} \times 3 \text{ route replicates} \times 20 \text{ realized paths} = 180$$

paired static-versus-DDDAS executions.

Because the same realized environment is shared across the three route replicates, these 180 runs do not constitute 180 independent environmental observations. There are instead

$$3 \times 20 = 60$$

unique day–realized-path environments. Recourse-search random seeds are distinct across route replicates so that the online optimizer itself does not reuse an identical random search trajectory.

Each DDDAS recourse attempt uses 100 conditional scenarios and the bounded stopping rule in Section 4.7. Static and DDDAS execution share the same complete realized environmental path. The static comparator simply continues executing its frozen preoperational route, whereas DDDAS may alter eligible residual suffixes using only causally observed information.

The realized composite criterion is

$$J_{\text{realized}} = D_{\text{realized}} + 2Z_{\text{realized}}, \quad (43)$$

and paired contrasts are defined as

$$\Delta J = J_{\text{static}} - J_{\text{DDDAS}}, \quad (44)$$

$$\Delta Z = Z_{\text{static}} - Z_{\text{DDDAS}}, \quad (45)$$

$$\Delta D = D_{\text{static}} - D_{\text{DDDAS}}. \quad (46)$$

Positive values favor DDDAS.

For scientific summaries, the three route replicates are first averaged within each fixed day–realized-path pair. This yields 20 Monte Carlo path-level contrasts per selected day. Within-day uncertainty is summarized by nonparametric bootstrap confidence intervals over these 20 realized environments, conditional on the frozen three-route ensemble.

Only three days are examined in this deep sequential experiment. We therefore do not use the 180 runs, or even the 60 path-level observations,

as independent cross-instance replicates. Cross-day R5-H summaries are descriptive rather than population-level hypothesis tests. Cross-instance evidence is instead provided by the nine-day preoperational experiment in Section 5.10.

No “realized CVaR” is constructed from an individual executed trajectory. The decision-time CVaR used by the online optimizer is a conditional distributional quantity, whereas each static-DDDAS comparison produces one subsequently realized outcome.

#### 5.12. *Performance measures*

The principal routing measures are:

1. total route distance  $D$ ;
2. mean realized total tardiness  $\mathbb{E}[Z]$ ;
3.  $\text{CVaR}_{0.95}(Z)$ ;
4. fraction of customers served on time;
5. maximum realized customer tardiness; and
6. the objective corresponding to the risk functional used in the relevant experiment.

Route-dependent mean and maximum OMI exposure are recorded as descriptive diagnostics. These quantities summarize the disruption states sampled at the actual departure times of the route and must not be interpreted as changes in the environmental Hawkes process itself.

For learning experiments, computational accounting includes online candidate-scenario evaluations, search runtime, and offline training cost. For sequential DDDAS, additional operational measures include the number of high-memory episodes, recourse attempts, executed route changes, evaluated non-null candidates, operator decisions, decision-cap terminations, conditional scenario evaluations, and recourse runtime.

A DDDAS trigger is therefore evaluated both as a computational event and as an operational decision. The opening of a high-memory episode does not imply that a route change will occur.

#### 5.13. *Paired analysis and inferential units*

The experimental design consistently distinguishes stochastic repetitions from independent operational instances.

For the Day 1 learning study, ten paired stochastic repetitions quantify source-instance variability. For zero-shot transfer, the target operational day

is the cross-instance unit ( $n = 8$ ). For the principal ALNS-Mean versus ALNS-CVaR comparison, the inferential unit is again the operational day ( $n = 9$ ), after averaging the three optimization replications within each day. For the deep DDDAS study, uncertainty is summarized separately within each of the three prospectively selected days, using the 20 realized environmental paths after averaging the three frozen route replicates.

This hierarchy avoids treating algorithmic seeds or common-random-number replications as independent real-world instances.

All reported method differences retain a fixed sign convention. For the nine-day risk experiment,

$$\text{ALNS-Mean} - \text{ALNS-CVaR} > 0$$

favors CVaR-oriented optimization. For DDDAS,

$$\text{static} - \text{DDDAS} > 0$$

favors dynamic recourse. For learned-versus-uniform transfer comparisons, the contrast is defined as

$$\text{learned} - \text{uniform},$$

so negative objective differences favor the learned policy.

#### *5.14. Computational environment and reproducibility*

All final computations were executed on a single Apple-silicon workstation with an Apple M5 Max processor, 18 CPU cores, and 64 GB of memory. The computational pipeline uses Python 3.12.13, NumPy 1.26.4, pandas 3.0.5, SciPy 1.13.1, PyTorch 2.2.2, and Google OR-Tools 9.10.4067.

The recurrent models are sufficiently small that GPU acceleration does not provide an advantage on the test platform; final learning and routing experiments therefore use CPU execution. Runtime comparisons are consequently made within a common hardware and software environment.

Scenario seeds, train/validation/search/test separation, route initializations, method configurations, and confirmatory sequential schedules are stored as reproducibility metadata. Common-random-number checks and sequential replay tests verify that static and dynamic methods observe identical realized environmental trajectories whenever a paired comparison requires them.

Table 2: Environmental diagnostics for the stationary-rate-matched persistence experiment. Each row is based on 10,000 simulated Hawkes trajectories. Increasing  $\eta$  changes temporal clustering while the stationary mean event rate is held fixed.

| $\eta$ | Fano factor | $P(N = 0)$ | $P(\tau_1 \leq 60)$ | Mean time-OMI |
|--------|-------------|------------|---------------------|---------------|
| 0.00   | 1.02        | 0.0003     | 0.6081              | 0.07550       |
| 0.15   | 1.40        | 0.0014     | 0.5432              | 0.06865       |
| 0.35   | 2.44        | 0.0052     | 0.4528              | 0.05832       |
| 0.55   | 4.77        | 0.0282     | 0.3437              | 0.04597       |
| 0.70   | 11.02       | 0.0917     | 0.2376              | 0.03543       |

## 6. Results

The results are organized around the four questions posed in Section 5. We first verify that the persistence manipulation changes the temporal organization of disruption events rather than merely their mean frequency. We then examine how this persistence affects routing risk under the mean-impact-matched control. Next, we assess the cross-instance value of explicit CVaR optimization and the incremental contribution of learned operator selection. Finally, we evaluate the causal sequential DDDAS mechanism on the three prospectively selected representative days.

### 6.1. Persistence reorganizes disruption exposure into intermittent bursts

The environmental simulation confirms that increasing the Hawkes branching ratio changes the temporal organization of disruptions substantially even when the stationary mean event intensity is held fixed through Eq. (38). Across 10,000 simulated environmental trajectories per regime, the mean number of events remained approximately constant, whereas event-count dispersion increased sharply with persistence.

Table 2 summarizes the principal diagnostics. The event-count Fano factor increased from approximately 1.02 at  $\eta = 0$  to 11.02 at  $\eta = 0.70$ . At the same time, the probability of observing no event over the operating horizon increased from 0.03% to 9.17%. The probability of an event occurring within the first 60 minutes decreased from 0.608 to 0.238.

The declining time-averaged OMI should not be interpreted as evidence that a more persistent environment is “safer.” Rather, higher persistence concentrates disruption activity into more strongly clustered episodes separated by longer quiet periods. Consistently with this interpretation, the upper tail

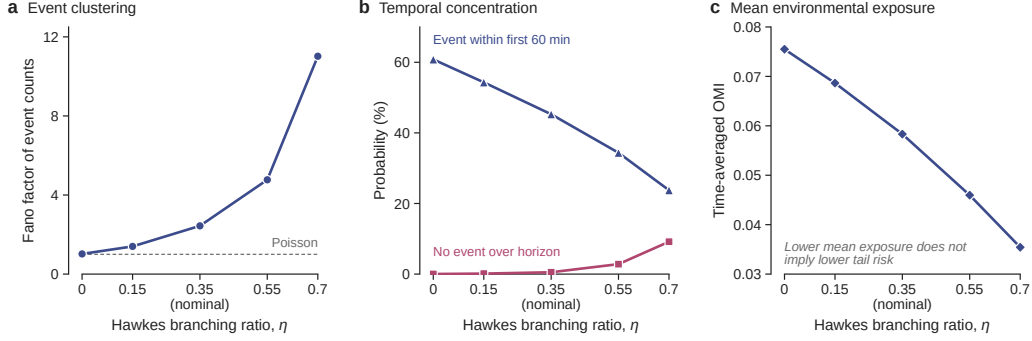


Figure 2: **Persistence reorganizes disruption activity under matched stationary event intensity.** (a) Event-count overdispersion rises sharply with the Hawkes branching ratio; the dashed line marks the Poisson reference (Fano factor = 1). (b) Higher persistence produces more event-free horizons and fewer early events, consistent with longer quiet intervals separated by clustered disruption episodes. (c) Time-averaged OMI decreases as exposure becomes more intermittent; this lower mean should not be interpreted as lower upper-tail routing risk. Environmental diagnostics use 10,000 trajectories per persistence regime.

of instantaneous OMI remains high across regimes, with its 0.99 quantile close to 0.94 throughout the persistence grid.

These environmental diagnostics are summarized in Fig. 2. Thus, the persistence manipulation primarily changes the *temporal structure* of environmental exposure: diffuse disruption activity at low  $\eta$  is replaced by increasingly intermittent and clustered bursts as  $\eta$  increases.

### 6.2. Persistence increases tail risk under matched mean environmental impact

The mean-impact-matched experiment isolates the operational consequences of this change in temporal structure. The impact coefficient  $\kappa_\eta$  is adjusted according to Eq. (40) so that the continuous-time mean environmental travel-time multiplier remains fixed across persistence regimes.

Despite this control, routing tail risk increases markedly with persistence. For the deterministic OR-Tools route, mean tardiness increases from 13.42 at  $\eta = 0$  to 17.19 at  $\eta = 0.70$ , while  $\text{CVaR}_{0.95}$  of total tardiness increases from 60.74 to 113.24. Across the ten paired repetitions, the increase in OR-Tools tail tardiness between  $\eta = 0$  and  $\eta = 0.70$  is

$$\Delta \text{CVaR}_{0.95} = 52.503, \quad 95\% \text{ CI} = [49.140, 55.932], \quad (47)$$

Table 3: Routing performance under the mean-environmental-impact-matched persistence control. Values are averages over ten paired repetitions.  $J_{\text{CVaR}} = D + 2\text{CVaR}_{0.95}(Z)$ .

| $\eta$ | Method    | Distance | Mean tardiness | CVaR <sub>0.95</sub> tardiness | $J_{\text{CVaR}}$ |
|--------|-----------|----------|----------------|--------------------------------|-------------------|
| 0.00   | OR-Tools  | 367.20   | 13.415         | 60.741                         | 488.682           |
|        | ALNS-Mean | 371.76   | 1.550          | 15.780                         | 403.320           |
|        | ALNS-CVaR | 376.62   | 0.599          | 8.228                          | 393.075           |
| 0.35   | OR-Tools  | 367.20   | 14.249         | 73.518                         | 514.236           |
|        | ALNS-Mean | 373.28   | 1.468          | 16.580                         | 406.441           |
|        | ALNS-CVaR | 378.22   | 0.801          | 8.701                          | 395.622           |
| 0.70   | OR-Tools  | 367.20   | 17.187         | 113.244                        | 593.688           |
|        | ALNS-Mean | 373.95   | 2.924          | 33.443                         | 440.836           |
|        | ALNS-CVaR | 378.21   | 1.872          | 21.612                         | 421.434           |

with all ten paired differences positive (exact two-sided sign-flip  $p = 0.00195$ ). Mean tardiness increases by 3.771 minutes (95% CI [3.408, 4.125],  $p = 0.00195$ ).

A detectable increase already occurs at the nominal persistence level: relative to  $\eta = 0$ , OR-Tools CVaR<sub>0.95</sub> increases by 12.777 at  $\eta = 0.35$ , again with all ten paired contrasts positive ( $p = 0.00195$ ).

Figure 3 visualizes the resulting upper-tail increase. Table 3 shows the corresponding aggregate performance of the three preoperational routing strategies at selected persistence levels.

The increase in tail risk cannot be attributed simply to a larger average environmental multiplier. For the unchanged OR-Tools route, for example, the route-sampled mean travel-time multiplier is approximately 1.0483 at  $\eta = 0$  and 1.0458 at  $\eta = 0.70$ , despite the sharp increase in CVaR<sub>0.95</sub>. This pattern implicates the temporal concentration of disruption exposure rather than a larger average route-level multiplier.

The relative advantage of CVaR-oriented search does not, however, increase robustly with persistence. Define

$$G(\eta) = J_{\text{CVaR}}(\text{ALNS-Mean}) - J_{\text{CVaR}}(\text{ALNS-CVaR}).$$

Although the point estimate increases between  $\eta = 0$  and  $\eta = 0.70$ , the paired interaction contrast is

$$G(0.70) - G(0) = 9.157, \quad 95\% \text{ CI} = [-15.230, 33.310], \quad (48)$$

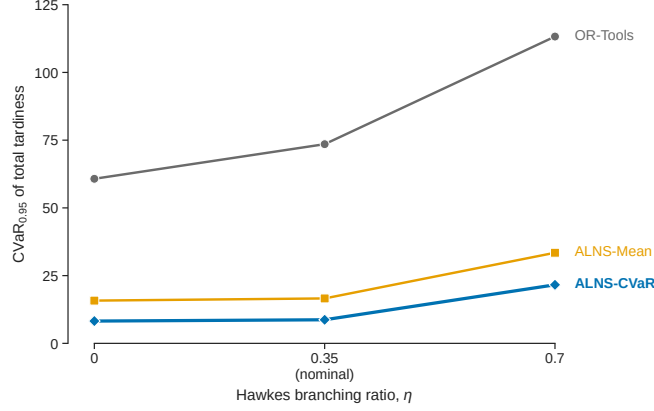


Figure 3: **Routing tail risk under the mean-environmental-impact-matched persistence control.** Upper-tail tardiness rises strongly with persistence even though the continuous-time mean environmental travel-time multiplier is matched across regimes. OR-Tools uses the same deterministic route across persistence levels; ALNS-Mean and ALNS-CVaR use the same adaptive search engine but optimize different risk functionals. Values are averages over ten paired repetitions.

with six positive and four negative paired differences ( $p = 0.494$ ). Persistence therefore produces a strong direct increase in tail risk, but the present data do not support the stronger claim that persistence systematically amplifies the *relative* benefit of CVaR-oriented search.

### 6.3. CVaR-oriented ALNS improves tail-risk performance across all nine days

The principal cross-instance comparison evaluates ALNS-Mean and ALNS-CVaR under the nominal disruption configuration on all nine Athens-derived operational days. The two methods use the same adaptive search engine, initial routes, neighbourhoods, stochastic search budget, and held-out test scenarios. Only the risk functional used during optimization differs.

The result is highly consistent across instances. Table 4 reports the day-level paired gains, after averaging the three optimization repetitions within each day. Positive values favor ALNS-CVaR.

For the tail-oriented objective,

$$\overline{\Delta J_{\text{CVaR}}} = 8.611, \quad (49)$$

with day-level standard deviation 4.628, median 10.545, and bootstrap 95%

Table 4: Day-level gains of ALNS-CVaR relative to ALNS-Mean under the nominal persistent-disruption configuration. Positive values favor CVaR-oriented optimization.

| Day | $\Delta J_{\text{CVaR}}$ | $\Delta \text{CVaR}_{0.95}(Z)$ |
|-----|--------------------------|--------------------------------|
| 1   | 10.545                   | 6.789                          |
| 2   | 4.960                    | 4.097                          |
| 3   | 1.832                    | 1.483                          |
| 4   | 11.609                   | 12.555                         |
| 5   | 1.773                    | 5.570                          |
| 6   | 8.968                    | 4.801                          |
| 7   | 10.856                   | 9.428                          |
| 8   | 13.562                   | 11.531                         |
| 9   | 13.396                   | 9.231                          |

confidence interval

$$[5.662, 11.314]. \quad (50)$$

All nine day-level contrasts are positive, giving an exact two-sided sign-flip  $p = 0.00391$ .

The same cross-instance consistency is observed directly for tail tardiness:

$$\overline{\Delta \text{CVaR}_{0.95}(Z)} = 7.276, \quad (51)$$

with standard deviation 3.663, median 6.789, and bootstrap 95% confidence interval

$$[5.027, 9.522]. \quad (52)$$

Again, all nine day-level contrasts favor ALNS-CVaR ( $p = 0.00391$ ).

Aggregating descriptively across the nine days, ALNS-CVaR attains mean distance 406.55, mean tardiness 1.92, tail tardiness 15.84, and on-time service fraction 0.9945. The corresponding ALNS-Mean values are 400.61, 3.24, 23.12, and 0.9911, respectively. Thus CVaR optimization accepts a modest increase in route distance—approximately 5.94 distance units on average—in exchange for lower mean and substantially lower upper-tail tardiness.

This trade-off is reflected in the aligned tail-risk objective: the average  $J_{\text{CVaR}}$  is 438.23 for ALNS-CVaR compared with 446.85 for ALNS-Mean. The deterministic OR-Tools route is shorter on average (398.87), but produces substantially greater mean tardiness (11.28) and tail tardiness (47.39), yielding average  $J_{\text{CVaR}} = 493.65$ .

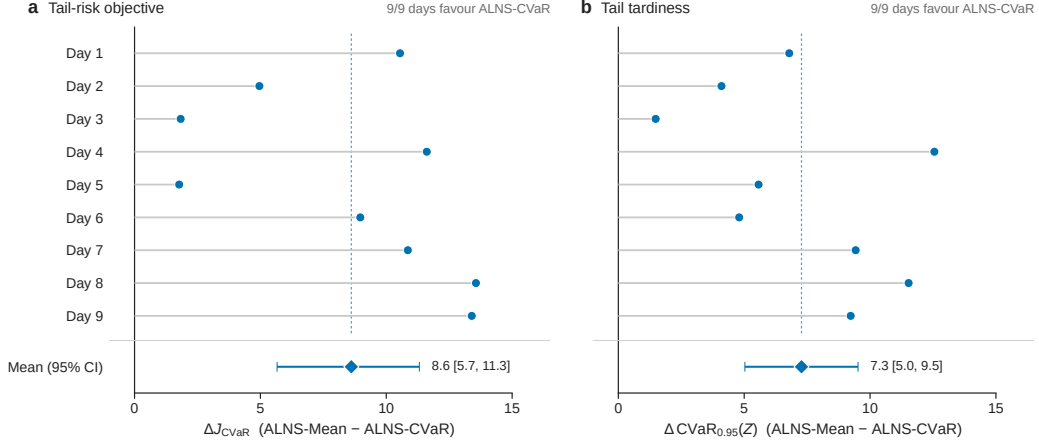


Figure 4: **Cross-instance gains from explicit CVaR-oriented planning.** Day-level contrasts compare ALNS-Mean with ALNS-CVaR after averaging the three paired optimization repetitions within each operational day; positive values favour CVaR-oriented search. (a) The aligned tail-risk objective improves on all nine days. (b)  $\text{CVaR}_{0.95}$  tardiness likewise improves on all nine days. The diamond and dashed vertical line report the across-day mean; the horizontal bar is the bootstrap 95% confidence interval.

Figure 4 shows the day-level consistency of both prespecified contrasts. The nine-day result is therefore not that CVaR optimization dominates every individual performance component. Rather, it consistently trades a small distance increase for a sufficiently large reduction in upper-tail service failure to improve the prespecified risk-oriented objective on every operational day.

#### 6.4. Learned operator selection does not provide a robust incremental advantage

The Day 1 learning experiment evaluates whether feed-forward or recurrent operator selection improves on the genuine adaptive ALNS backbone when the online stochastic search budget is held fixed.

Table 5 reports the ten-repetition source-day results. For mean-oriented optimization, the three methods produce closely overlapping performance. For CVaR-oriented optimization, the learned policies have somewhat lower point estimates of the aligned objective than ALNS-CVaR, but the between-method uncertainty remains substantial.

Figure 5 places these source-day point estimates beside the separate offline training burden. The paired contrasts do not establish a robust advantage of either learned policy over adaptive ALNS: the corresponding confidence

Table 5: Day 1 source-instance learning results over ten paired repetitions. Values are mean  $\pm$  standard deviation. The final column reports the objective aligned with the criterion being optimized:  $J_{\text{mean}}$  for mean-oriented rows and  $J_{\text{CVaR}}$  for CVaR-oriented rows.

| Criterion | Method | Mean tardiness    | CVaR <sub>0.95</sub> tardiness | Aligned objective    |
|-----------|--------|-------------------|--------------------------------|----------------------|
| Mean      | ALNS   | $1.440 \pm 1.017$ | $17.407 \pm 10.885$            | $374.970 \pm 3.240$  |
|           | RL-FF  | $1.783 \pm 1.489$ | $19.082 \pm 14.064$            | $375.585 \pm 3.629$  |
|           | RL-GRU | $1.355 \pm 0.951$ | $16.322 \pm 10.320$            | $375.479 \pm 3.225$  |
| CVaR      | ALNS   | $0.812 \pm 0.685$ | $9.413 \pm 5.879$              | $396.437 \pm 13.213$ |
|           | RL-FF  | $0.515 \pm 0.334$ | $7.785 \pm 3.995$              | $393.670 \pm 12.277$ |
|           | RL-GRU | $0.708 \pm 0.741$ | $8.501 \pm 7.323$              | $393.181 \pm 15.841$ |

intervals include zero. Likewise, learned operator selection does not significantly outperform the uniform operator-selection control under the matched search budget; the associated source-day comparisons all have  $p > 0.59$ .

The lack of a source-day significance result is reinforced by the zero-shot transfer experiment. Policies trained only on Day 1 are applied without fine-tuning to Days 2–9. Across the four prespecified learned-versus-uniform contrasts—feed-forward and GRU policies under mean- and CVaR-oriented objectives—none remains significant after Holm correction at the target-day level.

These results do not imply that learning-based operator selection is generally ineffective. They show instead that, under the present state representation, training protocol, and matched online-search budget, the learned selectors do not demonstrate a reproducible incremental advantage over either genuine adaptive ALNS or simple uniform operator selection.

The recurrent policy also fails to exhibit a consistent advantage over the feed-forward alternative. Thus, although the environment contains temporal persistence, recurrence itself does not guarantee improved search control once the relevant environmental and route-exposure summaries are already represented in the observable state.

This conclusion is especially relevant when computational cost is considered. Training one policy requires approximately  $2.3 \times 10^5$  scenario evaluations, around 9–10 times the stochastic evaluation cost of a single frozen-policy search. The offline learning cost therefore represents a substantial additional expenditure that is not justified by a corresponding robust performance gain in the present experiments.

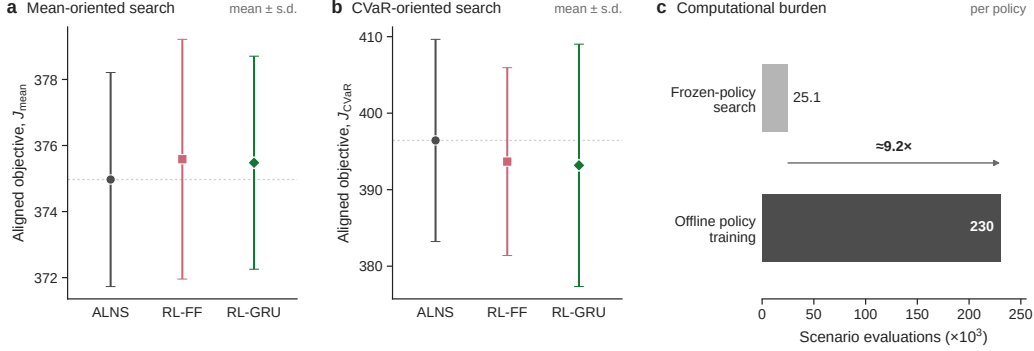


Figure 5: **Source-day performance and computational burden of learned operator selection.** (a,b) Points show the mean aligned objective and error bars one standard deviation over ten paired Day 1 repetitions; the dotted line marks the ALNS mean. These panels summarize descriptive source-day variability; the paired inferential comparisons reported in the text do not establish a robust learned-policy advantage. (c) Offline policy fitting requires approximately  $2.3 \times 10^5$  scenario evaluations per policy, about 9.2 times the stochastic evaluation budget of one frozen-policy search.

Table 6: Sequential DDDAS realized-performance contrasts. Values are means over 20 realized environmental paths after averaging the three frozen preoperational-route replicates within each path. Parentheses give path-bootstrap 95% confidence intervals. Positive values favor DDDAS.

| Day | $\Delta J_{\text{realized}}$ | $\Delta Z_{\text{realized}}$ | $\Delta D_{\text{realized}}$ |
|-----|------------------------------|------------------------------|------------------------------|
| 2   | -0.169 [-0.472, 0.182]       | 0.058 [0.000, 0.173]         | -0.285 [-0.505, -0.087]      |
| 5   | 0.087 [-0.377, 0.549]        | 0.056 [-0.131, 0.257]        | -0.025 [-0.300, 0.202]       |
| 7   | 0.122 [-1.481, 2.200]        | 0.416 [-0.298, 1.396]        | -0.710 [-1.100, -0.372]      |

### 6.5. Sequential DDDAS acts as selective rather than systematic recourse

The confirmatory DDDAS experiment evaluates sequential recourse on Days 2, 5, and 7 using 180 paired static-versus-DDDAS executions. These correspond to 60 unique day-environment combinations, with three frozen preoperational ALNS-CVaR route replicates evaluated against each realized environmental trajectory.

Table 6 reports the principal paired outcomes after first averaging the three route replicates within each day-realized-path unit. Positive differences favor DDDAS.

There is no systematic improvement in the realized composite objective. The mean of the three day-level  $\Delta J$  values is only 0.013, with Day 2 favoring

static execution and Days 5 and 7 showing small positive mean differences whose bootstrap intervals include zero.

Distance displays a clearer pattern. DDDAS increases realized distance on average on all three days:

$$\Delta D = -0.285, -0.025, -0.710$$

for Days 2, 5, and 7, respectively. For Days 2 and 7, the entire path-bootstrap interval is below zero. Mean tardiness moves in the opposite direction on all three days, but the estimated improvements are small relative to their uncertainty.

Day 7 illustrates the distinction particularly clearly. Its mean realized objective difference is slightly positive (0.122), whereas the median is negative ( $-0.033$ ). A leave-one-realized-path-out sensitivity analysis gives  $\Delta J$  means ranging from  $-0.717$  to  $0.575$ . The positive mean is therefore strongly influenced by a rare path on which DDDAS avoids substantial tardiness. No observation is removed from the reported analysis; the leave-one-out calculation is used only to characterize sensitivity of the descriptive mean.

#### 6.6. Most DDDAS triggers leave the incumbent route unchanged

The online mechanism is markedly conservative in its actual routing actions. Across the 180 sequential executions, the controller performs 523 recourse attempts but only 87 result in a route modification:

$$\frac{87}{523} = 16.6\%. \quad (53)$$

The remaining attempts retain the incumbent residual route.

At the run level, 115 of the 180 static-DDDAS comparisons contain no route change at all. For these 115 runs, static and DDDAS distance, tardiness, and realized objective are identical to numerical precision. Only 65 runs contain one or more actual route changes.

Table 7 summarizes the operational behavior by day.

The 25,000-operator-decision cap is reached in 194 attempts, or 37.1% of all recourse attempts. Only three of these capped searches ultimately produce a route change. Thus, in the great majority of cap-limited residual states, the bounded search exhausts its operator-decision budget without identifying a suffix that improves sufficiently on the incumbent conditional objective.

Table 7: Operational behavior of the bounded DDDAS controller.

| Day   | Runs | Episodes | Attempts | Changes | Cap terminations |
|-------|------|----------|----------|---------|------------------|
| 2     | 60   | 110      | 151      | 24      | 62               |
| 5     | 60   | 151      | 199      | 23      | 83               |
| 7     | 60   | 130      | 173      | 40      | 49               |
| Total | 180  | 391      | 523      | 87      | 194              |

Across all 523 recourse attempts, the sequential experiment performs 8,393,300 conditional scenario evaluations. Total measured DDDAS execution time over the 180 runs is 0.387 hours on the reported computational platform, corresponding to approximately 7.74 seconds per complete run on average. These timings reflect the optimized sequential implementation used for the confirmatory experiment.

The controller also frequently modifies a later part of the route rather than the immediately following arc. Only 19 of the 87 executed route modifications change the next customer visited at the decision epoch. The remaining changes preserve the immediate next node and modify a later portion of the residual suffix.

These observations reinforce the operational interpretation of the OMI trigger: crossing the threshold initiates *reconsideration* of the current route, not compulsory rerouting.

#### 6.7. Conditional risk improvement does not imply realized-cost improvement

The sequential logs permit direct comparison between the objective used when a rerouting decision is made and the subsequently realized outcome.

All 87 executed route modifications satisfy

$$G_{t_0} = \hat{J}_{\text{CVaR}}(r_{t_0}^{\text{inc}}) - \hat{J}_{\text{CVaR}}(r_{t_0}^{\text{sel}}) > 0, \quad (54)$$

so every executed change improves the local conditional CVaR objective on the decision-time scenario bank.

This consistency does not translate into guaranteed ex-post improvement. Among the 65 complete executions containing at least one route modification, the final realized composite objective favors DDDAS in 18 runs, is equal within a numerical tolerance of  $10^{-12}$  in 3 runs, and favors static execution in 44 runs.

Therefore,

$$G_{t_0} > 0 \not\Rightarrow J_{\text{DDDAS,realized}} < J_{\text{static,realized}}. \quad (55)$$

This is not a contradiction between the optimization and evaluation criteria. The first quantity compares two residual routes over a conditional distribution of possible future trajectories at the decision epoch. The second compares the two policies after one particular future trajectory has been realized. A route that reduces conditional tail risk can therefore incur a distance premium without receiving the corresponding insurance benefit on a specific realization.

Consistently with this interpretation, there is no stable descriptive relationship across the three days between the magnitude of the local conditional search gain and the final realized objective difference. The sequential results are therefore better characterized as evidence of *selective risk-sensitive recourse* than as evidence of systematic realized-cost improvement.

#### 6.8. Integrated interpretation

Figure 6 summarizes the realized DDDAS contrasts and its strong operational selectivity. Taken together, the experiments separate three effects that would otherwise be easy to conflate.

First, temporal persistence has an independent operational consequence: even after controlling the continuous-time mean environmental travel-time multiplier, greater clustering substantially increases upper-tail tardiness.

Second, explicitly accounting for this tail risk at the preoperational planning stage produces the most robust algorithmic improvement observed in the study. ALNS-CVaR improves both the aligned tail-risk objective and CVaR<sub>0.95</sub> tardiness relative to ALNS-Mean on all nine operational days.

Third, additional algorithmic complexity does not automatically yield further benefit. Learned operator selection does not robustly outperform adaptive or uniform operator selection under the matched evaluation protocol, and causal DDDAS recourse does not systematically reduce realized composite cost despite making locally risk-improving decisions whenever a route modification is executed.

The resulting empirical picture is therefore not one of uniform algorithmic dominance. Rather, the strongest evidence supports explicit *risk-aware preoperational planning*, while recurrent learning and online recourse provide adaptive mechanisms whose incremental value is more instance- and realization-dependent.

## 7. Discussion

The results support a differentiated view of routing under persistent disruptions. Temporal persistence matters, but its operational consequences do not arise simply because average travel times become larger. Explicit tail-risk optimization provides the clearest and most reproducible improvement across operational instances, whereas learned search control and online recourse provide more conditional benefits. This distinction is important because the three mechanisms address different stages of the decision process: the disruption model characterizes the environment, CVaR-oriented routing shapes the preoperational plan, and DDDAS provides state-dependent recourse during execution.

### 7.1. *Why persistence matters even when average impact is controlled*

The persistence experiments show that upper-tail service risk depends on the *temporal organization* of disruption, not only on its average magnitude. When the Hawkes branching ratio increases while the stationary mean event rate is held fixed, disruption activity becomes increasingly clustered: quiet periods become longer, but disruption episodes become more concentrated.

The stricter mean-impact-matched control reinforces this interpretation. Even after matching the continuous-time mean environmental travel-time multiplier across persistence regimes, tail tardiness increases substantially with  $\eta$ . For the unchanged OR-Tools route, for example,  $\text{CVaR}_{0.95}$  of tardiness rises from approximately 60.7 to 113.2 between  $\eta = 0$  and  $\eta = 0.70$ , even though the route-sampled mean travel-time multiplier does not increase.

This result illustrates why an average-delay representation can be insufficient for operational reliability analysis. Two disruption environments with similar mean impact may generate very different service outcomes if one distributes delay diffusely over time and the other concentrates it into bursts that arrive when route slack is already limited.

The mechanism should not be interpreted as feedback from the route into the Hawkes process. The disruption process remains exogenous. What changes is the interaction between a temporally clustered environment and the schedule recursion of the route. Once an early delay changes subsequent departure times, later arcs may encounter the same external disruption trajectory at less favorable points in the schedule. Persistence and schedule propagation can therefore reinforce one another operationally without constituting a route-to-environment feedback loop.

### 7.2. Tail-risk optimization is more robust than mean optimization

The most consistent algorithmic result is the advantage of explicitly optimizing the upper tail of tardiness. ALNS-CVaR improves both the aligned tail-risk objective and  $\text{CVaR}_{0.95}$  tardiness relative to ALNS-Mean on all nine operational days.

The gain is not free. CVaR-oriented solutions travel somewhat farther on average. This is precisely the trade-off expected from risk-sensitive routing: additional spatial cost is accepted in exchange for routes whose schedules are less exposed to severe service failures under unfavorable stochastic realizations.

Importantly, the evidence does not support the stronger claim that increasing persistence necessarily makes CVaR optimization progressively more valuable relative to mean optimization. Persistence strongly increases tail risk, but the interaction

$$G(0.70) - G(0)$$

is not statistically stable across paired repetitions. Thus, two conclusions should be separated:

persistence increases tail risk,

which is strongly supported, and

persistence necessarily amplifies the relative CVaR-search advantage,

which is not supported by the present data.

This distinction is relevant for practice because it suggests that tail-sensitive planning can be useful even before the environment reaches an extreme persistence regime. The value of CVaR does not depend on demonstrating a monotone interaction with the Hawkes branching ratio; it arises from protecting service performance against adverse realizations across heterogeneous operating days.

### 7.3. Learning does not automatically improve adaptive search

The learning experiments provide a useful counterpoint to the risk-optimization results. Neither the feed-forward nor recurrent policy demonstrates a robust incremental advantage over genuine adaptive ALNS, and

neither consistently outperforms a uniform operator-selection control under the matched online-search budget.

This result matters because temporal persistence can make recurrent learning architectures appear conceptually attractive. In the present setting, however, the observable state already contains stochastic tardiness summaries, capacity utilization, and route-dependent OMI exposure. A recurrent hidden state may therefore add limited information beyond these explicit state variables.

The zero-shot transfer experiment further indicates that operator-selection patterns learned on one operational day are not automatically portable to other days. This does not imply that reinforcement learning is unsuitable for vehicle-routing search. Rather, it shows that the added complexity of learning requires empirical justification under controlled baselines, especially when offline training cost is substantial.

This qualification is particularly important in applied routing systems. A learned component should not be regarded as operationally superior merely because it is more sophisticated than an adaptive heuristic. If a trained policy does not provide stable gains beyond simpler controls, then the additional training, maintenance, and validation burden becomes part of the deployment decision.

#### *7.4. Sequential recourse: conditional improvement versus realized outcome*

The DDDAS experiment reveals a second distinction that is central to risk-sensitive online decision making.

Every executed route modification improves the local conditional search objective on the decision-time scenario bank. Nevertheless, among complete runs containing at least one route change, realized composite cost improves in only a minority of cases.

This apparent tension is a natural consequence of stochastic optimization. At a decision time  $t_0$ , the controller compares residual routes over a conditional distribution of possible futures. The selected route is preferable with respect to this distributional criterion. Execution then reveals only one future trajectory. A decision that reduces conditional tail risk may therefore incur additional distance in a realization where the adverse future against which it provided protection does not occur.

In this sense, online CVaR recourse resembles an insurance decision more than a deterministic shortest-path correction. The decision may be rational

ex ante even when its insurance premium—typically additional distance—is not repaid in the subsequently realized trajectory.

The Day 7 results illustrate this mechanism particularly clearly. The mean realized objective difference is slightly favorable to DDDAS, but the median is slightly unfavorable and the mean is sensitive to a rare high-consequence trajectory in which substantial tardiness is avoided. Such events are precisely the type of outcome that tail-sensitive control is designed to protect against, but they also imply that average realized gains can be highly variable in finite samples.

The operational logs reinforce the selective nature of the controller. Only 16.6% of recourse attempts result in a route modification, and approximately 37.1% of attempts reach the bounded operator-decision limit. Moreover, most executed changes do not alter the immediately following customer. The DDDAS layer should therefore be interpreted as a mechanism for selective residual-plan reconsideration rather than continual reactive rerouting.

### 7.5. Managerial implications

Several managerial implications follow from these results.

First, route planning under persistent disruption should not be based solely on average travel-time inflation. The persistence experiments show that two operating environments with comparable mean travel-time impact can generate substantially different tail-risk profiles. Managers concerned with service reliability should therefore monitor not only the magnitude of congestion or delay, but also its temporal concentration and persistence.

Second, the strongest operational evidence favors incorporating tail risk into the *preoperational* routing decision. The nine-day comparison shows that CVaR-oriented ALNS consistently sacrifices a limited amount of distance to reduce severe tardiness. For delivery systems in which late service carries disproportionate contractual, reputational, or downstream operational cost, this trade-off may be preferable to minimizing expected tardiness or route length alone.

Third, online replanning should not be interpreted as an obligation to change a route whenever conditions deteriorate. In the confirmatory DDDAS experiment, most triggers do not lead to a route modification. This is operationally desirable: unnecessary rerouting can create driver confusion, invalidate customer communications, increase dispatch workload, and consume computational resources without improving service.

A useful implementation principle is therefore

detect  $\rightarrow$  evaluate  $\rightarrow$  change only if justified,

rather than

detect  $\rightarrow$  reroute automatically.

The DDDAS architecture supports precisely this distinction.

### 7.6. *Implications for human-in-the-loop planning*

The selective behavior of the online controller also suggests a natural role for human-in-the-loop decision support.

In many delivery operations, dispatchers and drivers retain responsibility for route changes because optimization systems do not observe every operational constraint. Temporary road restrictions, parking availability, customer preferences, driver familiarity, and informal local knowledge may not be represented explicitly in the routing model.

A memory-aware DDDAS layer can therefore be used as a *decision-support filter* rather than as a fully autonomous rerouting system. A high-OMI episode can flag that the current schedule is operating under elevated persistent disruption, while conditional suffix optimization can quantify whether an alternative route appears materially preferable.

This creates at least three possible levels of automation:

1. *advisory mode*: the system reports elevated disruption state and candidate alternatives to a dispatcher;
2. *confirmation mode*: the system proposes a reroute only when the conditional risk improvement exceeds a specified operational threshold, but a human approves execution;
3. *autonomous mode*: residual routes are modified automatically under preauthorized operating rules.

The results favor a cautious progression across these levels. Because a conditional improvement does not guarantee a lower realized cost, decision interfaces should communicate the nature of the gain explicitly. For example, presenting a recommendation as “lower predicted tail risk” is more informative than presenting it as “better route.”

This distinction may improve operator trust by aligning the system’s recommendation with the uncertainty actually being optimized.

### 7.7. Strategic implications for service-level agreements

The results also have implications for service-level agreements (SLAs) and contract design.

Many logistics contracts are framed around thresholds such as delivery within a time window, maximum allowed delay, or penalties applied once lateness exceeds a specified level. Such contracts are intrinsically tail-sensitive even when the underlying routing system is optimized for average cost.

A mean-oriented routing policy can therefore be misaligned with the economic structure of the service agreement. Two routing plans with similar expected tardiness may have very different probabilities of severe delay or very different upper-tail losses.

CVaR provides a natural operational bridge between these two levels. Although it is not itself an SLA metric, optimizing  $\text{CVaR}_{0.95}$  explicitly penalizes the subset of scenarios in which service degradation is most severe. This can be particularly relevant when lateness cost is nonlinear or when a small number of severe failures dominate customer dissatisfaction.

The nine-day results suggest that this form of protection can be obtained with a moderate distance premium. The practical question for an operator is therefore not simply whether the CVaR route is longer, but whether the additional routing cost is smaller than the expected economic consequence of severe SLA failure.

This interpretation also suggests that the objective weight  $\lambda_T$  should ultimately be linked to business-specific service economics rather than treated as a universal constant. The present experiments use a fixed common value to enable controlled algorithmic comparison; deployment would require calibration against actual penalty structures and service priorities.

### 7.8. Memory-aware decision support as an anticipatory paradigm

The concept of memory-aware decision support remains useful, but its meaning should be stated carefully.

In the present model, “memory” does not imply that realized vehicle delays feed back into the disruption process, nor does it imply an intrinsically infinite-dimensional non-Markovian system. For the exponential Hawkes kernel, the relevant disruption history is summarized by the current excitation state  $m(t)$ .

The anticipatory value comes instead from using that state to distinguish two environments that may look similar instantaneously but have different recent histories and therefore different near-term event intensity.

A conventional reactive system asks:

What is the current travel condition?

A memory-aware system additionally asks:

What does the recent disruption history imply about  
the current persistence state?

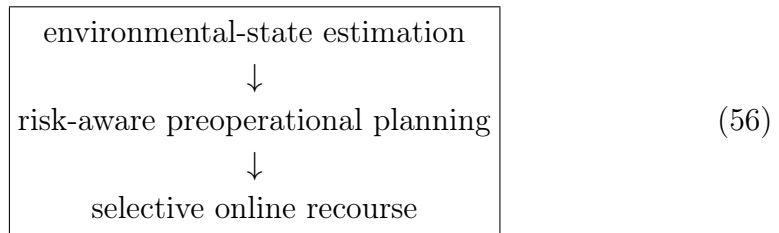
This shift is modest mathematically but potentially important operationally. The OMI is not a forecast that a specific future incident will occur. Rather, it is a compact state variable that summarizes how strongly recent events contribute to present disruption intensity.

The resulting decision paradigm is therefore *anticipatory but not clairvoyant*. It uses observed history to update the distribution of possible futures while preserving causal information constraints.

This distinction is especially relevant for DDDAS. The sequential controller does not inspect the realized future trajectory. At each departure epoch, it reconstructs a conditional future from the observed event prefix and completed travel times. The controller can therefore adapt to the evolving disruption state without relying on retrospective information.

#### 7.9. Implications for routing-system architecture

Taken together, the experiments suggest a layered architecture for disruption-prone last-mile routing:



The first layer estimates the current persistent disruption state. The second constructs a route that is robust to adverse service outcomes before vehicles depart. The third reevaluates selected residual routes only when newly observed information makes reconsideration worthwhile.

The empirical evidence in this study is strongest for the second layer. CVaR-oriented planning provides the most reproducible improvement across

instances. The third layer contributes flexibility, but its incremental realized value is more variable and must be balanced against computational and organizational cost.

This architecture also helps clarify the potential role of learning. RL need not replace the optimization layer. It can instead be introduced only where repeated operational experience demonstrates that learned search control adds value beyond adaptive heuristics. The present results indicate that such an advantage should be established empirically rather than assumed.

#### *7.10. Limitations and scope of inference*

The study has several limitations that delimit the interpretation of the results.

First, the operational instances are derived from real Athens delivery data, but the disruption process is experimental. The Hawkes parameters are not estimated from a historical database of Athens traffic incidents. The persistence experiments should therefore be interpreted as controlled stress tests rather than empirical estimates of disruption dynamics in Athens.

Second, the tightened time windows and vehicle capacities are experimental design choices. They provide a common regime in which service reliability can be compared, but they are not claimed to reproduce the contractual or fleet configuration of the original operator.

Third, the Hawkes model is univariate and spatially homogeneous at the environment level. Real urban disruption is likely to depend on location, road class, time of day, weather, incident type, and network interactions. A multivariate or spatially structured point process would provide a richer representation but would also require correspondingly richer calibration data.

Fourth, the DDDAS optimizer performs local recourse on the residual suffix of the triggered vehicle while preserving customer assignment. It does not solve a full fleet-wide dynamic reassignment problem. Consequently, the online results quantify the value of computationally bounded local recourse rather than the upper bound obtainable from unrestricted dynamic reoptimization.

Fifth, the deep sequential DDDAS experiment uses three prospectively selected representative days. The 20 realized paths per day characterize stochastic within-instance variability, but they do not create additional independent operational instances. Cross-instance conclusions therefore rely primarily on the nine-day preoperational analysis.

Sixth, the learning results are conditional on the present state representation, policy architectures, and training protocol. Failure to observe a robust RL advantage should not be generalized to all learning-assisted vehicle-routing methods.

Finally, the mean-impact-matched persistence control equalizes the continuous-time mean environmental multiplier, not the route-sampled multiplier for every method. Because route departure times differ, small differences in route-specific exposure remain possible and are part of the operational interaction between routing and the external disruption trajectory.

These limitations point to several natural extensions, including spatial or multivariate disruption models, empirical calibration of persistence from incident or probe-vehicle data, heterogeneous fleets and service agreements, fleet-wide dynamic reassignment, and deployment studies in which dispatcher responses to risk-sensitive recommendations can be observed directly.

## 8. Conclusions

This study examined last-mile routing under temporally persistent disruptions by separating three distinct decision layers: the external disruption process, risk-aware preoperational planning, and causal online recourse.

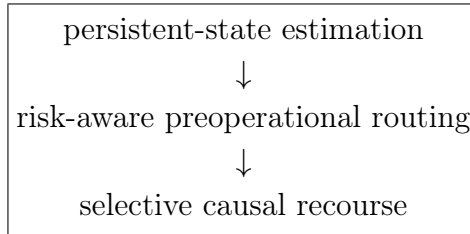
The first main result is that disruption persistence matters beyond average travel-time inflation. When the Hawkes branching ratio is increased while the stationary mean event rate is controlled, disruption activity becomes more temporally clustered. Under the stricter control that also matches the continuous-time mean environmental travel-time multiplier, upper-tail tardiness still increases substantially. Thus, routing reliability depends not only on the average magnitude of disruption but also on how disruption is organized over time.

The second and strongest algorithmic result concerns preoperational risk-aware planning. Across all nine Athens-derived operational days, CVaR-oriented ALNS consistently improves both the aligned tail-risk objective and  $\text{CVaR}_{0.95}$  tardiness relative to mean-oriented ALNS. This improvement comes with a moderate increase in route distance, illustrating a clear reliability–efficiency trade-off rather than dominance across every individual performance measure. The results therefore support explicit tail-risk optimization when severe service delays carry disproportionate operational or contractual consequences.

By contrast, additional learning complexity does not yield a robust incremental advantage in the present experiments. Feed-forward and recurrent operator-selection policies do not consistently outperform genuine adaptive ALNS or a uniform operator-selection control under matched online-search budgets, and zero-shot transfer to other operating days does not reveal a stable generalization benefit. Recurrence is therefore not automatically valuable simply because the disruption environment contains temporal persistence.

The sequential DDDAS experiment provides a complementary result. On-line recourse is highly selective: most disruption-triggered reconsiderations retain the incumbent residual route. Every executed route modification improves the local conditional risk objective at the decision epoch, but this conditional improvement does not imply lower realized cost on the single trajectory that subsequently occurs. The DDDAS layer should therefore be interpreted as bounded risk-sensitive recourse rather than as a mechanism that guarantees ex-post cost improvement.

Taken together, the findings support a layered view of decision support under persistent disruption:



Among these layers, the most robust empirical evidence in the present study supports explicit CVaR-based planning. Learning-assisted search and dynamic replanning remain potentially useful extensions, but their incremental value is more dependent on instance structure, available information, and computational budget.

Future work should focus on empirically calibrated spatial or multivariate disruption processes, heterogeneous fleets and service-level costs, richer fleet-wide dynamic reassignment, and deployment settings in which dispatcher or driver responses can be incorporated explicitly. Such extensions would allow the memory-aware framework developed here to move from controlled operational stress testing toward fully data-calibrated dynamic decision support.

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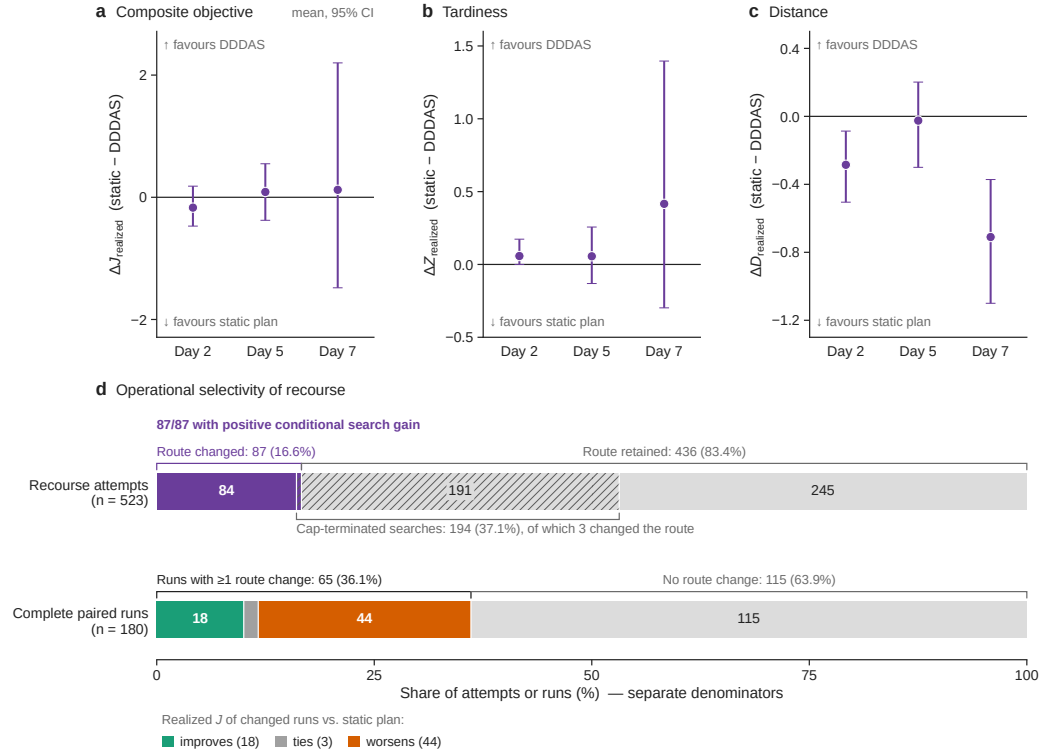
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**Algorithm 1** Causal memory-triggered DDDAS recourse

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1: Deploy a preoperational route set  $x$ 
2: Initialize the Hawkes process and arm the episode trigger
3: while unserved customers remain do
4:   Select the next vehicle departure in chronological order
5:   Complete any waiting and service at the current node
6:   Set current departure time  $t_0$ 
7:   Observe disruption events with  $\tau_q < t_0$ 
8:   Compute  $m(t_0)$  and  $\text{OMI}(t_0)$ 
9:   if  $\text{OMI}(t_0) < 0.55$  then
10:     Rearm the trigger
11:   end if
12:   if trigger is armed and  $\text{OMI}(t_0) \geq 0.55$  then
13:     Open a high-memory episode and disarm the trigger
14:   end if
15:   if vehicle is eligible for recourse in the current episode then
16:     Construct a conditional scenario bank from observed information
17:     Run bounded local ALNS-CVaR on the vehicle's residual suffix
18:     if conditional objective improves then
19:       Replace the unexecuted suffix
20:     end if
21:     Mark the vehicle as attempted for the current episode
22:   end if
23:   Start the next arc using  $1 + \kappa \text{OMI}(t_0)$ 
24:   Hold that arc multiplier fixed until arrival
25:   Register the realized baseline OD time as observed upon arrival
26: end while
27: Compute full-fleet realized distance, tardiness, and service outcomes
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**Figure 6: Realized effects and operational selectivity of causal DDDAS recourse.** Static-minus-DDDAS contrasts are shown for (a) realized composite objective, (b) realized tardiness and (c) realized distance; positive values favour DDDAS. Points are means over 20 realized environmental paths after averaging the three frozen route replicates; error bars are path-bootstrap 95% confidence intervals. (d) Attempt-level and run-level outcomes are shown as proportions of their own denominators. Only 87 of 523 recourse attempts modify a route, and 115 of 180 complete runs contain no route change. Every executed route modification improves the local conditional search objective, but realized full-run cost does not improve systematically: among the 65 changed runs, 18 improve, 3 tie within  $10^{-12}$ , and 44 worsen relative to static execution.